02.

Decimal Fractions

Type - 9

1. Find the greatest among these fractions.
5/6, 6/11, 2/3, 8/9, 6/7
(a) 2/3 (b) 8/9
(c) 5/6 (d) 6/7
RRB JE - 01/06/2019 (Shift-III)

Ans. (b) From question :-
$\frac{5}{6} = 0.83$
$\frac{6}{11} = 0.54$
$\frac{2}{3} = 0.67$
$\frac{8}{9} = 0.89$
$\frac{6}{7} = 0.85$
Hence, the greatest fraction is $0.89 = \frac{8}{9}$

2. Find the difference between the greatest and the least fraction among 2/3, 3/4, 4/5, 5/6.
(a) 3/5 (b) 1/7 (c) 1/6 (d) 2/5
RRB JE - 22/05/2019 (Shift-I)

Ans : (c) $\frac{2}{3} = 0.66$ the least fraction,
$\frac{3}{4} = 0.75$
$\frac{4}{5} = 0.8$
$\frac{5}{6} = 0.83$ the greatest fraction,
Hence, the required difference $= \frac{5}{6} - \frac{2}{3} = \frac{5-4}{6} = \frac{1}{6}$

3. Which of the following is the greatest?
(a) 15/16 (b) 24/25
(c) 34/35 (d) 19/20
RRB JE - 02/06/2019 (Shift-II)

Ans. (c) 15/16 = 0.937
24/25 = 0.96
34/35 = 0.97 (the greatest fraction)
19/20 = 0.95

4. Find the greatest among these fractions.
5/11, 3/15, 12/11, 4/7, 9/12
(a) 12/11 (b) 3/15
(c) 9/12 (d) 4/7
RRB RPF-SI -10/01/2019 (Shift-I)

Ans : (a)
$\frac{5}{11} = 0.45,\ \frac{3}{15} = 0.2,\ \frac{12}{11} = 1.09,\ \frac{4}{7} = 0.57,\ \frac{9}{12} = 0.75$
Hence, the required largest fraction will be $= \frac{12}{11}$

5. Find the least among these fractions.
$\frac{1}{10}, \frac{1}{100}, \frac{9}{1000}, \frac{500}{10000}$
(a) $\frac{500}{10000}$ (b) $\frac{1}{100}$
(c) $\frac{1}{10}$ (d) $\frac{9}{1000}$
RRB RPF Constable -17/01/2019 (Shift-I)

Ans. (d) : $\frac{1}{10} = 0.1$
$\frac{1}{100} = 0.01$
$\frac{9}{1000} = 0.009$
$\frac{500}{10000} = 0.05$
$0.1 > 0.05 > 0.01 > 0.009$
Hence, it is clear that the required fraction is $\frac{9}{1000}$.

6. Which of the following fractions is the greatest?
(a) 8/19 (b) 9/22
(c) 10/23 (d) 11/24
RRB RPF-SI -11/01/2019 (Shift-III)

Ans : (d) From options
$\frac{8}{19} = 0.421, \frac{9}{22} = 0.409, \frac{10}{23} = 0.43, \frac{11}{24} = 0.458$
Hence, the required greatest fraction is $\frac{11}{24}$.

7. Arrange the following ratios in decreasing order, which number will be the last?
11:14, 17:21, 5:7, 2:3
(a) 17:21 (b) 5:7
(c) 2:3 (d) 11:14
RRB Group-D – 05/10/2018 (Shift-II)

Ans. (c) The given proportional numbers are-
$\frac{11}{14}, \frac{17}{21}, \frac{5}{7}, \frac{2}{3} = \frac{33}{42}, \frac{34}{42}, \frac{30}{42}, \frac{28}{42}$
The decreasing order of the numbers,
$\frac{17}{21} > \frac{11}{14} > \frac{5}{7} > \frac{2}{3}$
Hence, the least proportional number will be 2/3.

8. Which of the following fractions is the largest?

$\frac{1}{8}, \frac{2}{12}, \frac{3}{16}, \frac{4}{20}$

(a) $\frac{3}{16}$ (b) $\frac{4}{20}$

(c) $\frac{1}{8}$ (d) $\frac{2}{12}$

RRB Group-D – 12/10/2018 (Shift-III)

Ans : (b) $\frac{1}{8} = 0.125$

$\frac{2}{12} = 0.166$

$\frac{3}{16} = 0.187$

$\frac{4}{20} = 0.200$

Hence, it is clear that $\frac{4}{20}$ is the greatest fraction.

9. Which of the following fractions is the largest?

$\frac{3}{15}, \frac{5}{20}, \frac{8}{64}, \frac{25}{1000}$

(a) $\frac{5}{20}$ (b) $\frac{8}{64}$

(c) $\frac{3}{15}$ (d) $\frac{25}{1000}$

RRB Group-D – 10/10/2018 (Shift-III)

Ans : (a)

(a) $\frac{5}{20} = 0.25$ (b) $\frac{8}{64} = 0.125$

(c) $\frac{3}{15} = 0.2$ (d) $\frac{25}{1000} = 0.025$

Hence, the largest fraction is $\frac{5}{20}$.

10. Which of the following is the smallest fraction?

$\frac{4}{9}, \frac{5}{4}, \frac{3}{8}, \frac{6}{7}$

(a) $\frac{3}{8}$ (b) $\frac{4}{9}$

(c) $\frac{6}{7}$ (d) $\frac{5}{4}$

RRB Group-D – 16/10/2018 (Shift-II)

Ans : (a)

$\frac{4}{9} = 0.444$

$\frac{5}{4} = 1.25$

$\boxed{\frac{3}{8} = 0.375}$

$\frac{6}{7} = 0.857$

Hence, it is clear that $\frac{3}{8}$ is the smallest fraction.

11. Which of the following fractions is the largest?

(a) $\frac{29}{77}$ (b) $\frac{8}{21}$

(c) $\frac{5}{14}$ (d) $\frac{25}{66}$

RRB Group-D – 16/10/2018 (Shift-II)

Ans : (b) $\frac{29}{77} = 0.376$

$\boxed{\frac{8}{21} = 0.380}$

$\frac{5}{14} = 0.357$, $\frac{25}{66} = 0.378$

Hence, it is clear that $\frac{8}{21}$ is the largest fraction.

12. Which of the following is the smallest fraction?

$\frac{6}{11}, \frac{13}{18}, \frac{15}{22}, \frac{19}{36}, \frac{5}{6}$

(a) $\frac{19}{36}$ (b) $\frac{13}{18}$

(c) $\frac{6}{11}$ (d) $\frac{5}{6}$

RRB Group-D – 24/10/2018 (Shift-II)

Ans. (a) : $\frac{6}{11}, \frac{13}{18}, \frac{15}{22}, \frac{19}{36}, \frac{5}{6}$

$\frac{6}{11} = 0.54$, $\frac{13}{18} = 0.72$

$\frac{15}{22} = 0.68$, $\frac{19}{36} = 0.52$

$\frac{5}{6} = 0.83$

Hence, it is clear that the smallest fraction is $\frac{19}{36}$.

13. $\frac{8}{6}, \frac{6}{4}, \frac{5}{3}, \frac{9}{5}$

The largest fraction among the given fractions is:

(a) $\frac{5}{3}$ (b) $\frac{6}{4}$

(c) $\frac{9}{5}$ (d) $\frac{8}{6}$

RRB Paramedical Exam – 20/07/2018 (Shift-III)

Ans. (c) : $\frac{8}{6} = 1.33$, $\frac{6}{4} = 1.50$

$\frac{5}{3} = 1.66$, $\frac{9}{5} = 1.80$

Hence, it is clear that the required largest fraction is $\frac{9}{5}$.

14. Which of the following is the smallest fraction?

$\frac{3}{15}, \frac{5}{20}, \frac{8}{64}, \frac{25}{1000}$

(a) $\frac{8}{64}$ (b) $\frac{25}{1000}$

(c) $\frac{5}{20}$ (d) $\frac{3}{15}$

RRB Group-D – 15/10/2018 (Shift-II)

Ans : (b)

$\frac{3}{15} = 0.2, \quad \frac{5}{20} = 0.25$

$\frac{8}{64} = 0.125, \quad \frac{25}{1000} = 0.025$

Hence, the required smallest fraction is $\frac{25}{1000}$.

15. Which of the following fractions is the least of all?

3/4, 3/5, 3/8, 3/11

(a) 3/4 (b) 3/7

(c) 3/11 (d) 3/8

RRB NTPC 31.03.2016 Shift : 3

Ans : (c) $\frac{3}{4}, \frac{3}{5}, \frac{3}{8}, \frac{3}{11}$

$\because$ The numerators of all the fractions are same, so the fraction of the greatest denominator will be the least of all.

Hence, the required fraction will be $\frac{3}{11}$.

16. Which of the following fractions is the least of all?

(a) 6/5 (b) 4/3

(c) 3/2 (d) 5/4

RRB NTPC 29.03.2016 Shift : 3

Ans : (a)

$\frac{6}{5} = 1.2, \quad \frac{4}{3} = 1.33$

$\frac{3}{2} = 1.5, \quad \frac{5}{4} = 1.25$

Hence, it is clear that $\frac{6}{5}$ is the required least fraction of all.

17. Which of the following fractions is largest?

(a) 3/4 (b) 4/5

(c) 5/6 (d) 7/8

RRB NTPC 27.04.2016 Shift : 1

Ans : (d) $\frac{3}{4} = 0.75$, $\frac{4}{5} = 0.80$, $\frac{5}{6} = 0.833$, $\frac{7}{8} = 0.875$

Hence, the required largest fraction is $\frac{7}{8}$.

18. The smallest of the fractions among 5/8, 3/4, 13/16, 7/12 is_____.

(a) 5/8 (b) 3/4

(c) 13/16 (d) 7/12

RRB NTPC 27.04.2016 Shift : 2

Ans : (d)

$\frac{5}{8} = 0.625, \quad \frac{3}{4} = 0.75, \quad \frac{13}{16} = 0.8125$

$\frac{7}{12} = 0.58$

Hence, the smallest fraction is $\frac{7}{12}$.

19. Which of the following fractions is the least?

(a) $\frac{3}{4}$ (b) $\frac{4}{5}$

(c) $\frac{5}{6}$ (d) $\frac{6}{7}$

RRB NTPC 29.04.2016 Shift : 3

Ans. (a) : From options,

$\frac{3}{4} = 0.75, \quad \frac{4}{5} = 0.80$

$\frac{5}{6} = 0.83, \quad \frac{6}{7} = 0.85$

Hence, the required least fraction is $\frac{3}{4}$.

20. Which of the following is the least fraction?

(a) 3/5 (b) 5/6

(c) 2/3 (d) 4/5

RRB NTPC 30.04.2016 Shift : 2

Ans : (a) $\frac{3}{5} = 0.6, \frac{5}{6} = 0.83, \frac{2}{3} = 0.66, \frac{4}{5} = 0.8$

Hence, it is clear that the least fraction will be $\frac{3}{5}$.

21. Which of the following is a reducible fraction?

(a) $\frac{91}{15}$ (b) $\frac{79}{26}$

(c) $\frac{105}{112}$ (d) $\frac{41}{17}$

RRB ALP & Tec. (09-08-18 Shift-II)

Ans : (c) A fraction whose fractional shares have a common factor, is called reducible fraction.

$\frac{105}{112} = \frac{7 \times 15}{7 \times 16}$

Type - 2

22. Arrange the following fractions in descending order.

5/6, 3/7, 8/9, 3/14

(a) 8/9, 5/6, 3/7, 3/14 (b) 8/9, 3/14, 3/7, 5/6

(c) 5/6, 8/9, 3/7, 3/14 (d) 3/7, 8/9, 5/6, 3/14

RRB JE - 22/05/2019 (Shift-III)

Ans : (a)

$\frac{5}{6} = 0.83, \quad \frac{3}{7} = 0.42, \quad \frac{8}{9} = 0.88, \quad \frac{3}{14} = 0.21$

Hence, the descending order of the fractions is $\frac{8}{9}, \frac{5}{6}, \frac{3}{7}, \frac{3}{14}$.

23. Which of the following fractions are in descending order?

(a) 5/8, 7/12, 3/4, 13/16
(b) 7/12, 13/16, 3/4, 5/8
(c) 5/8, 7/12, 13/16, 3/4
(d) 13/16, 3/4, 5/8, 7/12

RRB JE - 26/06/2019 (Shift-III)

Ans : (d) Arranging in descending order-
(a) 5/8 = 0.62, 7/12 = 0.58, 3/4 = 0.75, 13/16 = 0.81
(b) 7/12 = 0.58, 13/16 = 0.81, 3/4 = 0.75, 5/8 = 0.62
(c) 5/8 = 0.62, 7/12 = 0.58, 13/16 = 0.81, 3/4 = 0.75
(d) 13/16 = 0.81, 3/4 = 0.75, 5/8 = 0.62, 7/12 = 0.58
Hence, option (d) is in descending order.

24. Which of the following fractions are in ascending order?

(a) 0.65, 0.76, 0.67, 0.86,
(b) 0.65, 0.86, 0.67, 0.76
(c) 0.65, 0.67, 0.76, 0.86
(d) 0.67, 0.65, 0.76, 0.86

RRB RPF Constable -17/01/2019 (Shift-III)

Ans : (c)
$0.65 < 0.67 < 0.76 < 0.86$
Hence, option (c) is in ascending order.

25. Which of the following is true for given numbers?

(a) 13/33<32/47<20/47<25/27
(b) 13/33<20/47<25/27<32/27
(c) 13/33<20/47<32/47<25/27
(d) 20/47<13/33<32/47<25/27

RRB RPF-SI -13/01/2019 (Shift-I)

Ans : (c) The given fractions–

$$\frac{13}{33} = 0.\overline{39},\ \frac{20}{47} = 0.42$$

$$\frac{32}{47} = 0.68,\ \frac{25}{27} = 0.92$$

$$0.\overline{39} < 0.42 < 0.68 < 0.92$$

Hence, $\frac{13}{33} < \frac{20}{42} < \frac{32}{47} < \frac{25}{27}$

26. Which from the following is in descending order?

(a) 2/3, 3/4, 4/5, 1/2
(b) 3/4, 4/5, 1/2, 2/3
(c) 4/5, 3/4, 2/3, 1/2
(d) 4/5, 1/2, 2/3, 3/4

RRB RPF Constable -25/01/2019 (Shift-I)

Ans : (c) From option (c),

$\frac{4}{5}, \frac{3}{4}, \frac{2}{3}, \frac{1}{2}$

$\because$ LCM of the denominators = 60
Hence, again from option (c),

$$= \frac{48}{60} > \frac{45}{60} > \frac{40}{60} > \frac{30}{60}$$

Hence, $\frac{4}{5}, \frac{3}{4}, \frac{2}{3}, \frac{1}{2}$ are in descending order.

27. Whose ascending order from the following numbers is correct?

(a) $\frac{5}{6}, \frac{3}{5}, \frac{7}{9}$
(b) $\frac{3}{5}, \frac{5}{6}, \frac{7}{9}$
(c) $\frac{3}{5}, \frac{7}{9}, \frac{5}{6}$
(d) $\frac{7}{9}, \frac{3}{5}, \frac{5}{6}$

RRB RPF Constable -20/01/2019 (Shift-I)

Ans : (c) The given fractions,

$\frac{5}{6} = 0.83$ $\quad$ $\frac{3}{5} = 0.6$

$\frac{7}{9} = 0.77$

The required ascending order is 0.6, 0.77, 0.83

$$= \frac{3}{5}, \frac{7}{9}, \frac{5}{6}$$

28. Find the value of $\sqrt{2}$ to eight digits of decimal.

(a) 1.41421356
(b) 1.41421354
(c) 1.41421346
(d) 1.41421366

RRB Group-D – 12/10/2018 (Shift-II)

Ans : (a) Extracting the square root of $2.\overline{00}\,\overline{00}\,\overline{00}\,\overline{00}\,\overline{00}\,\overline{00}\,\overline{00}\,\overline{00}$ by division method, the result of 8-digit number 1.41421356, is obtained.
That is $\sqrt{2} = 1.41421356$

29. Which among the following is the correct ascending order of the numbers?

(a) $\frac{1}{3}, \frac{4}{15}, 0.33$
(b) $\frac{1}{3}, 0.33, \frac{4}{15}$
(c) $\frac{4}{15}, 0.33, \frac{1}{3}$
(d) $0.33, \frac{4}{15}, \frac{1}{3}$

RRB NTPC 17.01.2017 Shift-1

Ans : (c) From the given fractions,

$\frac{1}{3} = 0.333, \frac{4}{15} = 0.266$ and 0.33

$0.266 < 0.33 < 0.333$

$\frac{4}{15} < 0.33 < \frac{1}{3}$

Hence, the required ascending order of the numbers will be $\frac{4}{15}, 0.33, \frac{1}{3}$.

30. The correct ascending order of the given numbers is:

(a) $\frac{3}{10}, \frac{9}{15}, \frac{1}{3}$
(b) $\frac{4}{15}, \frac{1}{3}, \frac{1}{10}$
(c) $\frac{1}{3}, \frac{3}{10}, \frac{4}{15}$
(d) $\frac{4}{15}, \frac{3}{10}, \frac{1}{3}$

RRB NTPC 17.01.2017 Shift-3

Ans : (d) $\frac{3}{10} = 0.30, \frac{4}{15} = 0.26, \frac{1}{3} = 0.33$

$0.26 < 0.30 < 0.33$

Hence, the required ascending order is $\frac{4}{15} < \frac{3}{10} < \frac{1}{3}$.

31. Which of the following is correct for the given numbers?
(a) 12/43<32/67<45/81<22/55
(b) 12/43<22/55<45/81<32/67
(c) 12/43<32/67<22/55<45/81
(d) 12/43<22/55<32/67<45/81

RRB NTPC 04.04.2016 Shift : 3

Ans : (d) From options,

$$\frac{12}{43}=0.27,\ \frac{32}{67}=0.47,\ \frac{45}{81}=0.55,\ \frac{22}{55}=0.4$$

$$\frac{12}{43}<\frac{22}{55}<\frac{32}{67}<\frac{45}{81}$$

Hence, $0.27 < 0.4 < 0.47 < 0.55$

32. Which one is correct for the following given numbers?
(a) 25/51 < 12/19 < 47/63 < 63/79
(b) 25/51 < 63/79 < 12/19 < 47/63
(c) 12/19 < 25/51 < 63/79 < 47/63
(d) 63/79 < 47/63 < 12/19 < 25/51

RRB NTPC 03.04.2016 Shift : 3

Ans : (a)

$$\frac{25}{51}=0.490 \qquad \frac{12}{19}=0.631$$

$$\frac{47}{63}=0.746 \qquad \frac{63}{79}=0.797$$

The correct order is 0.490 < 0.631 < 0.746 < 0.797

Hence, $\frac{25}{51}<\frac{12}{19}<\frac{47}{63}<\frac{63}{79}$

33. Which of the following is correct for the given numbers?
(a) 13/21 < 57/97 < 52/94 < 36/79
(b) 36/79 < 57/97 < 52/94 < 13/21
(c) 36/79 < 52/94 < 13/21 < 57/97
(d) 36/79 < 52/94 < 57/97 < 13/21

RRB NTPC 02.04.2016 Shift : 1

Ans : (d) $\frac{13}{21}=0.619,\qquad \frac{57}{97}=0.587$

$$\frac{52}{94}=0.553,\qquad \frac{36}{79}=0.455$$

Hence, $\frac{36}{79}<\frac{52}{94}<\frac{57}{97}<\frac{13}{21}$

34. Which of the following is correct for the given numbers?
(a) 3/8 < 19/73 < 29/47 < 17/39
(b) 19/73 < 3/8 < 17/39 < 29/47
(c) 19/73 < 3/8 < 29/47 < 17 / 39
(d) 19/73 < 29/47 < 3/8 < 17/39

RRB NTPC 02.04.2016 Shift : 2

Ans : (b) From the given fractions,

$$\frac{3}{8}=0.37$$

$$\frac{19}{73}=0.26$$

$$\frac{29}{47}=0.61$$

$$\frac{17}{39}=0.43$$

Hence, the correct order is $\frac{19}{73}<\frac{3}{8}<\frac{17}{39}<\frac{29}{47}$

35. Which of the following is in ascending order?
(a) 2/3, 3/4, 4/5, 1/2
(b) 3/3, 4/5, 1/2, 2/3
(c) 1/2, 2/3, 3/4, 4/5
(d) 4/5, 1/2, 2/3, 3/4

RRB NTPC 30.03.2016 Shift : 1

Ans : (c) Solving the options,

(a) $\frac{2}{3},\frac{3}{4},\frac{4}{5},\frac{1}{2}\rightarrow 0.67, 0.75, 0.8, 0.5$

(b) $\frac{3}{4},\frac{4}{5},\frac{1}{2},\frac{2}{3}\rightarrow 0.75, 0.8, 0.5, 0.67$

(c) $\frac{1}{2},\frac{2}{3},\frac{3}{4},\frac{4}{5}\rightarrow 0.5, 0.67, 0.75, 0.8$

(d) $\frac{4}{5},\frac{1}{2},\frac{2}{3},\frac{3}{4}\rightarrow 0.8, 0.5, 0.67, 0.75$

Hence option (c) is an ascending order of the given fractions.

36. Arrange the following fractions in correct descending order.
(a) 7/9, 11/47, 5/13, 24/29
(b) 24/29, 11/47, 7/9, 5/13
(c) 11/47, 5/13, 7/9, 24/29
(d) 24/29, 7/9, 5/13, 11/47

RRB NTPC 18.01.2017 Shift : 3

Ans : (d) The given fractions,

$$\frac{24}{29}=0.827,\qquad \frac{5}{13}=0.384$$

$$\frac{7}{9}=0.777,\qquad \frac{11}{47}=0.234$$

Hence, arranging into descending order-

$$\frac{24}{29},\frac{7}{9},\frac{5}{13},\frac{11}{47}$$

37. Which is the correct ascending order of the given numbers?
(a) $\frac{2}{3},\frac{5}{6},\frac{3}{4}$ (b) $\frac{3}{4},\frac{2}{3},\frac{5}{6}$
(c) $\frac{2}{3},\frac{3}{4},\frac{5}{6}$ (d) $\frac{5}{6},\frac{3}{4},\frac{2}{3}$

RRB NTPC 11.04.2016 Shift : 1

Ans : (c) From the given fractions,

$$\frac{2}{3}=0.66.....$$

$$\frac{5}{6}=0.83.....$$

$$\frac{3}{4}=0.75.....$$

Arranging into ascending order- 0.66 < 0.75 < 0.83

$$\Rightarrow \frac{2}{3},\frac{3}{4},\frac{5}{6}$$

38. Which is the correct ascending order of the given numbers?

(a) 1/3, 1/5, 2/5 (b) 1/3, 2/5, 1/5

(c) 1/5, 1/3, 2/5 (d) 1/5, 2/5, 1/3

RRB NTPC 11.04.2016 Shift : 2

Ans : (c) $\frac{1}{3} = 0.33$

$\frac{1}{5} = 0.2$, $\frac{2}{5} = 0.4$

Hence, the required ascending order is 0.2, 0.3, 0.4

$= \frac{1}{5}, \frac{1}{3}, \frac{2}{5}$

39. Whose ascending order is correct from the given fractions?

(a) $\frac{5}{8}, \frac{19}{24}, \frac{11}{16}$ (b) $\frac{11}{16}, \frac{5}{8}, \frac{19}{24}$

(c) $\frac{5}{8}, \frac{11}{16}, \frac{19}{24}$ (d) $\frac{19}{24}, \frac{11}{16}, \frac{5}{8}$

RRB NTPC 11.04.2016 Shift : 3

Ans : (c) $\frac{5}{8} = 0.625$ $\frac{19}{24} = 0.791$ $\frac{11}{16} = 0.687$

Hence, the required ascending order is 0.625 < 0.687 < 0.791

$\Rightarrow \frac{5}{8} < \frac{11}{16} < \frac{19}{24}$

40. Which one is correct for the following given numbers?

(a) 13/41 < 32/67<45/81<23/53

(b) 13/41 < 23/53 < 45/81 < 32/67

(c) 13/41<32/67 < 23/53 < 45/81

(d) 13/41 < 23/53 < 32/67 < 45/81

RRB NTPC 10.04.2016 Shift : 3

Ans : (d) From given fractions,

$\frac{13}{41} = 0.31$, $\frac{32}{67} = 0.47$

$\frac{45}{81} = 0.55$, $\frac{23}{53} = 0.43$

0.31 < 0.43 < 0.47 < 0.55

$\Rightarrow \frac{13}{41} < \frac{23}{53} < \frac{32}{67} < \frac{45}{81}$

41. Arrange the following fractions in ascending order.

(a) 11/17 < 41/63 < 3/7 <21/29

(b) 41/63 < 3/7 < 21/29 <11/17

(c) 3/7 < 11/17 < 41/63 < 21/29

(d) 3/7 < 11/17 < 21/29 < 41/63

RRB NTPC 09.04.2016 Shift : 3

Ans : (c) $\frac{11}{17} = 0.64, \frac{41}{63} = 0.65$

$\frac{3}{7} = 0.42, \frac{21}{29} = 0.72$

Hence, the required ascending order is 0.42, 0.64, 0.65,

$0.72 = \frac{3}{7} < \frac{11}{17} < \frac{41}{63} < \frac{21}{29}$

42. Which of the following given numbers is in ascending order?

(a) $\frac{1}{3}, 0.3, \frac{2}{8}$ (b) $0.3, \frac{1}{3}, \frac{2}{8}$

(c) $\frac{2}{8}, 0.3, \frac{1}{3}$ (d) $\frac{1}{3}, \frac{2}{8}, 0.3$

RRB NTPC 07.04.2016 Shift : 2

Ans : (c) From given fractions,

$\frac{1}{3} = 0.33, \frac{2}{8} = 0.25$ and 0.3

Arranging in ascending order-

$0.25 < 0.3 < 0.33 = \frac{2}{8}, 0.3, \frac{1}{3}$

43. What is the correct ascending order of the following numbers?

$\frac{3}{4}, \frac{17}{24}, \frac{2}{3}$

(a) $\frac{17}{24}, \frac{3}{4}, \frac{2}{3}$ (b) $\frac{3}{4}, \frac{2}{3}, \frac{17}{24}$

(c) $\frac{2}{3}, \frac{3}{4}, \frac{17}{24}$ (d) $\frac{2}{3}, \frac{17}{24}, \frac{3}{4}$

RRB NTPC 19.01.2017 Shift : 1

Ans : (d)

$\frac{3}{4} = 0.75$, $\frac{17}{24} = 0.708\overline{3}$, $\frac{2}{3} = 0.\overline{6}$

Hence, the correct ascending order is $\frac{2}{3}, \frac{17}{24}, \frac{3}{4}$.

44. Which is the correct ascending order of the given numbers?

(a) $\frac{3}{7}, 0.3, \frac{2}{7}$ (b) $0.3, \frac{2}{7}, \frac{3}{7}$

(c) $\frac{2}{7}, 0.3, \frac{3}{7}$ (d) $\frac{2}{7}, \frac{3}{7}, 0.3$

RRB NTPC 22.04.2016 Shift : 2

Ans : (c) Denoting the given fractions in decimals,

$0.3, \frac{3}{7} = 0.42, \frac{2}{7} = 0.28$

Hence, the required ascending order is 0.28, 0.3, 0.42 = $\frac{2}{7}, 0.3, \frac{3}{7}$

45. Which is the correct ascending order of the given numbers?

(a) $\frac{1}{2}, \frac{2}{3}, \frac{7}{12}$ (b) $\frac{7}{12}, \frac{2}{3}, \frac{1}{2}$

(c) $\frac{1}{2}, \frac{7}{12}, \frac{2}{3}$ (d) $\frac{2}{3}, \frac{1}{2}, \frac{7}{12}$

RRB NTPC 26.04.2016 Shift : 2

Ans : (c) Denoting the given fractions in decimals,

$\frac{1}{2} = 0.5$, $\frac{2}{3} = 0.66$, $\frac{7}{12} = 0.58$

Hence, the correct ascending order is 0.5, 0.58, 0.66

$= \frac{1}{2} < \frac{7}{12} < \frac{2}{3}$.

46. Which is the correct ascending order of the given numbers?

(a) $\frac{1}{3},\frac{3}{4},\frac{5}{8}$ (b) $\frac{5}{8},\frac{3}{4},\frac{1}{3}$

(c) $\frac{1}{3},\frac{5}{8},\frac{3}{4}$ (d) $\frac{3}{4},\frac{1}{3},\frac{5}{8}$

RRB NTPC 26.04.2016 Shift : 3

Ans : (c) $\frac{1}{3}=0.33,\ \frac{3}{4}=0.75,\ \frac{5}{8}=0.625$

Hence, the required ascending order is $\frac{1}{3},\frac{5}{8},\frac{3}{4}$.

47. What is the correct ascending order for the given fractions?

(a) 22/7, 13/17, 11/19, 2/3
(b) 11/19, 2/3, 13/17, 22/7
(c) 2/3, 11/19, 13/17, 22/7
(d) 2/3, 13/17, 11/19, 22/7

RRB NTPC 28.04.2016 Shift : 2

Ans : (b) $\frac{22}{7}=3.14,\ \frac{13}{17}=0.76$

$\frac{11}{19}=0.57,\ \frac{2}{3}=0.67$

Hence, the ascending order will be $\frac{11}{19}<\frac{2}{3}<\frac{13}{17}<\frac{22}{7}$

48. Write the correct ascending order of the given fractions.

(a) 3/7, 15/41, 19/35, 7/11
(b) 15/41, 3/7, 19/35, 7/11
(c) 3/7, 15/41, 7/11, 19/35
(d) 19/35, 7/11, 15/41, 3/7

RRB NTPC 28.04.2016 Shift : 3

Ans : (b) From options,

$\because \frac{15}{41}=0.365,\quad \frac{3}{7}=0.428$

$\frac{19}{35}=0.542,\quad \frac{7}{11}=0.636$

Hence, the ascending order is $\frac{15}{41}<\frac{3}{7}<\frac{19}{35}<\frac{7}{11}$.

49. Which is the correct ascending order of the given numbers?

(a) $\frac{5}{6},\frac{11}{12},\frac{8}{9}$ (b) $\frac{8}{9},\frac{5}{6},\frac{11}{12}$

(c) $\frac{5}{6},\frac{8}{9},\frac{11}{12}$ (d) $\frac{11}{12},\frac{8}{9},\frac{5}{6}$

RRB NTPC 30.04.2016 Shift : 3

Ans : (c) from options,

$\frac{5}{6}=0.83$

$\frac{11}{12}=0.92$

$\frac{8}{9}=0.88$

Hence, the required ascending order of the numbers is $\frac{5}{6},\frac{8}{9},\frac{11}{12}$.

Type - 3

50. Which of the following is terminating decimal?

(a) $\frac{1}{32}$ (b) $\frac{1}{24}$

(c) $\frac{1}{96}$ (d) $\frac{1}{48}$

RRB RPF-SI -05/01/2019 (Shift-III)

Ans : (a) From options,

$\frac{1}{32}=0.03125$

$\frac{1}{24}=0.416666667$

$\frac{1}{96}=0.0104166667$

$\frac{1}{48}=0.0208333333$

If the number obtained by dividing a number, the digits do not repeat over and over. We call it a terminating decimal.

$\therefore\ \frac{1}{32}=0.03125$

Hence, option (a) $\frac{1}{32}$ is in terminating decimal.

51. Which of the following will have a terminating decimal expansion?

(a) $\frac{57}{120}$ (b) $\frac{47}{150}$

(c) $\frac{61}{110}$ (d) $\frac{43}{140}$

RRB Group-D – 20/09/2018 (Shift-II)

Ans : (a) From options,

(a) $\frac{57}{120}=0.475$

(b) $\frac{47}{150}=0.31\overline{3}$

(c) $\frac{61}{110}=0.55\overline{45}$

(d) $\frac{43}{140}=0.30\overline{714285}$

Hence, $\frac{57}{120}$ will have a terminating decimal expansion.

52. Which of the following numbers will have a value of terminating decimal?

(a) $\frac{9}{45}$ (b) $\frac{6}{45}$

(c) $\frac{3}{45}$ (d) $\frac{12}{45}$

RRB Group-D – 24/09/2018 (Shift-II)

Ans : (a) $\frac{9}{45} = 0.2$

$\frac{6}{45} = 0.1333$

$\frac{3}{45} = 0.06667$

$\frac{12}{45} = 0.26667$

Hence, it is clear that the value of option (a) $\frac{9}{45}$, is a terminating decimal.

53. Which of the following will give expiration decimal?

(a) $\frac{12}{72}$ (b) $\frac{6}{72}$

(c) $\frac{9}{72}$ (d) $\frac{3}{72}$

RRB Group-D – 03/10/2018 (Shift-III)

Ans : (c) $\frac{12}{72} = \frac{1}{6} = 0.166666---$

$\frac{6}{72} = \frac{1}{12} = 0.083333---$

$\frac{9}{72} = \frac{1}{8} = 0.125$

$\frac{3}{72} = \frac{1}{24} = 0.041666----$

Hence, option (c) $\frac{9}{72} = \frac{1}{8} = 0.125$ is correct.

54. Which of the following fractions will not have a value in recurring decimal?

(a) $\frac{20}{56}$ (b) $\frac{25}{56}$

(c) $\frac{10}{56}$ (d) $\frac{21}{56}$

RRB Group-D – 20/09/2018 (Shift-I)

Ans. (d) : (a) $\frac{20}{56} = 0.357142$ (b) $\frac{25}{56} = 0.44642$

(c) $\frac{10}{56} = 0.178571$ (d) $\frac{21}{56} = 0.375$

Hence, the value of option (d) $\frac{21}{56} = 0.375$ is not in recurring decimal.

55. Which of the following options is an example of recurring decimal?

(a) $\frac{24}{60}$ (b) $\frac{24}{90}$

(c) $\frac{24}{120}$ (d) $\frac{24}{30}$

RRB Group-D – 17/09/2018 (Shift-II)

Ans : (b) (a) $\frac{24}{60} = 0.4$ (b) $\frac{24}{90} = 0.266$

(c) $\frac{24}{120} = 0.2$ (d) $\frac{24}{30} = 0.8$

Hence, option (b) $\frac{24}{90}$ is an example of recurring decimal.

56. Which of the following vulgar fractions, when written as a decimal and its value will not be found in a terminating decimal?

(a) $\frac{27}{480}$ (b) $\frac{21}{640}$

(c) $\frac{81}{450}$ (d) $\frac{240}{450}$

RRB Group-D – 22/10/2018 (Shift-III)

Ans : (d) (a) $\frac{27}{480} = 0.05625$

(b) $\frac{21}{640} = 0.0328125$

(c) $\frac{81}{450} = 0.18$

(d) $\frac{240}{450} = 0.5333 = 0.5\overline{3}$

Hence, it is clear that option (d) $\frac{240}{450} = 0.5333 = 0.5\overline{3}$ is not giving a value in terminating decimal.

57. Which of the following fractions will be a terminating decimal?

(a) $\frac{6}{144}$ (b) $\frac{12}{144}$

(c) $\frac{3}{144}$ (d) $\frac{9}{144}$

RRB Group-D – 03/12/2018 (Shift-II)

Ans : (d) $\frac{6}{144} = \frac{3}{72} = \frac{1}{24} = 0.041\overline{6}$

$\frac{12}{144} = \frac{1}{12} = 0.08\overline{3}$

$\frac{3}{144} = \frac{1}{48} = 0.0208\overline{3}$

$\frac{9}{144} = \frac{1}{16} = 0.0625$

Hence, it is clear that option (d) $\frac{9}{144} = \frac{1}{16} = 0.0625$ is a terminating decimal fraction.

58. Which of the following is terminating decimal expansion?

(a) $\frac{1}{6}$ (b) $\frac{17}{25}$

(c) $\frac{10}{3}$ (d) $\frac{1}{11}$

RRB NTPC 17.01.2017 Shift-2

Ans : (b) (a) $\frac{1}{6} = 0.166$ ……. (Non-terminating)

(b) $\frac{17}{25} = 0.68$ …….. (Terminating)

(c) $\frac{10}{3} = 3.33$ …….. (Non-terminating)

(d) $\frac{1}{11} = 0.090909$ … (Non-terminating)

Hence, option (b) is terminating decimal.

59. Which of the following will yield a recurring decimal?

(a) $\frac{21}{30}$ (b) $\frac{21}{120}$

(c) $\frac{21}{60}$ (d) $\frac{21}{90}$

RRB ALP & Tec. (20-08-18 Shift-II)

Ans : (d) $\frac{21}{30} = 0.7$, $\frac{21}{120} = 0.175$, $\frac{21}{60} = 0.35$

$\frac{21}{90} = 0.23333 \Rightarrow 0.2\overline{3}$

Hence option (d) is recurring decimal.

60. Which of the following will give a terminating decimal?

(a) $\frac{3}{36}$ (b) $\frac{12}{36}$

(c) $\frac{9}{36}$ (d) $\frac{6}{36}$

RRB ALP & Tec. (10-08-18 Shift-III)

Ans : (c) From option (c),

$$\frac{9}{36} = \frac{1}{4} = 0.25$$

Hence, it is clear that option (c) is a fraction of terminating decimal.

61. The fraction from the ones listed below that will not lead to a recurring decimal is:

(a) $\frac{8}{56}$ (b) $\frac{6}{56}$

(c) $\frac{4}{56}$ (d) $\frac{7}{56}$

RRB ALP & Tec. (09-08-18 Shift-III)

Ans : (d) From options,

(a) $\frac{8}{56} = 0.142857.....$

(b) $\frac{6}{56} = 0.107142.....$

(c) $\frac{4}{56} = 0.071428.....$

(d) $\frac{7}{56} = 0.125$

Type - 3

62. Evaluate:

$0.6\overline{23}$

(a) $\frac{623}{999}$ (b) $6\frac{23}{999}$

(c) $\frac{617}{990}$ (d) $6\frac{23}{990}$

RRB JE - 31/05/2019 (Shift-II)

Ans : (c) $0.6\overline{23}$

$= \frac{623-6}{990}$ (From decimal fraction system)

$= \frac{617}{990}$

63. Find the value of $0.1\overline{8}$.

(a) 27/90 (b) 17/90

(c) 17/100 (d) 18/100

RRB JE - 29/05/2019 (Shift-III)

Ans : (b) $0.1\overline{8}$

$= \frac{18-1}{10\times 9} = \frac{17}{90}$

64. What is the correct expression of $0.06\overline{54}$ [(¯) sign represents continuous decimal)]?

(a) $\frac{18}{275}$ (b) $\frac{18}{277}$

(c) 654 (d) $\frac{654}{1000}$

RRB RPF-SI -10/01/2019 (Shift-III)

Ans : (a) $0.06\overline{54}$

$= \frac{654-6}{9900} = \frac{648}{9900} = \frac{18}{275}$

65. $0.\overline{047619}$, when written as a vulgar fraction, is equal to-

(a) $\frac{1}{21}$ (b) $\frac{1}{19}$

(c) $\frac{1}{23}$ (d) $\frac{1}{17}$

RRB Group-D – 19/09/2018 (Shift-II)

Ans. (a) : From given number,

$$0.\overline{047619} = \frac{047619}{999999} = \frac{1}{21}$$

66. Convert $0.\overline{6}$ into fraction:

(a) $\frac{6}{3}$ (b) $\frac{2}{3}$

(c) $\frac{2}{6}$ (d) $\frac{4}{3}$

RRB Group-D – 28/09/2018 (Shift-II)

Ans. (b) : Given,

$0.\overline{6} = \frac{6}{9} = \frac{2}{3}$, …….. (From decimal fraction system)

67. Which of the following fractions' result will not be a recurring decimal?

(a) $\frac{10}{30}$ (b) $\frac{12}{30}$

(c) $\frac{14}{30}$ (d) $\frac{8}{30}$

RRB Paramedical Exam – 21/07/2018 (Shift-III)

Ans : (b) From options,

(a) $\frac{10}{30} = \frac{1}{3} = 0.\overline{3}$

(b) $\frac{12}{30} = \frac{4}{10} = \frac{2}{5} = 0.4$

(c) $\frac{14}{30} = \frac{7}{15} = 0.4\overline{6}$

(d) $\frac{8}{30} = \frac{4}{15} = 0.2\overline{6}$

Hence, it is clear that option (b) is not in recurring decimal.

68. Which of the following vulgar fractions will not end when represented in decimal?

(a) $\frac{81}{150}$ (b) $\frac{80}{150}$

(c) $\frac{15}{48}$ (d) $\frac{21}{600}$

RRB Group-D – 10/10/2018 (Shift-II)

Ans : (b) From options,

(a) $\frac{81}{150} = 0.54$

(b) $\frac{80}{150} = 0.5333333$

(c) $\frac{15}{48} = 0.3125$

(d) $\frac{21}{600} = 0.035$

Hence, it is clear that option (b) is not and decimal.

69. $0.02\overline{36}$ when written as the simplest form, what will be the obtained vulgar fraction?

(a) $\frac{13}{1100}$ (b) $\frac{13}{9999}$

(c) $\frac{13}{3300}$ (d) $\frac{13}{550}$

RRB Group-D – 25/09/2018 (Shift-I)

Ans : (d) $0.02\overline{36}$

$0.02\overline{36} = \frac{0.2\overline{36}}{10}$

$= \frac{236-2}{10\times 990} = \frac{234}{9900}$

$= \frac{117}{4950} = \frac{13}{550}$

70. Express $\frac{7}{11}$ in the form of decimal.

(a) $0.\overline{623}$ (b) $0.\overline{633}$

(c) $0.\overline{63}$ (d) $0.\overline{62}$

RRB Group-D – 25/09/2018 (Shift-II)

Ans : (c) The decimal of $\frac{7}{11}$

```
      .6363
11)70
   66
   ---
    40
    33
    ---
     70
     66
     ---
      40
      33
      --
       7
```

Hence, in 0.6363, two digits of the number (6 and 3) are repeating itself.

71. Show $0.08\overline{36}$ in the form of vulgar fraction.

(a) $\frac{46}{555}$ (b) $\frac{23}{1100}$

(c) $\frac{23}{275}$ (d) $\frac{828}{9900}$

RRB Group-D – 15/11/2018 (Shift-III)

Ans : (c) : $0.08\overline{36} = ?$

$= \frac{836-8}{9900}$

$= \frac{828}{9900} = \frac{23}{275}$

72. $\frac{1}{450}$, when written as recurring decimal, will be equal to:

(a) 0.2 (b) 0.02

(c) 0.002 (d) 0.0002

RRB Group-D – 12/10/2018 (Shift-I)

Ans. (c) : Given, $\frac{1}{450}$

On dividing,

```
      0.0022
450)  1000
       900
       ----
       1000
        900
        ---
        100
```

Hence, $\frac{1}{450} = 0.00\overline{2}$

73. Express $0.09\overline{87}$ as a vulgar fraction in its lowest form?

(a) $\frac{163}{1650}$ (b) $\frac{329}{9990}$

(c) $\frac{326}{3300}$ (d) $\frac{163}{1665}$

RRB Group-D – 08/10/2018 (Shift-III)

Ans : (c) Let x = $0.09\overline{87}$

$100x = 9.\overline{87}$ Multiplying by 100 in both side,

= 9.8787 .....

= 9.78 + 0.098787 .....

100x = 9.78 + x

99x = 9.78

$x = \frac{9.78}{99}$

$= \frac{978}{9900} = \frac{326}{3300}$

74. Which of the following fractions will yield a recurring decimal?

(a) $\frac{27}{60}$ (b) $\frac{27}{72}$

(c) $\frac{27}{48}$ (d) $\frac{27}{84}$

RRB Group-D – 05/10/2018 (Shift-III)

Ans. (d) (a) $\frac{27}{60} = 0.45$ (b) $\frac{27}{72} = 0.375$

(c) $\frac{27}{48} = 0.5625$ (d) $\frac{27}{84} = 0.32\overline{142857}$

Hence, the fraction $\frac{27}{84}$, gives a recurring decimal.

75. If $0.\overline{41}$ is expressed as the vulgar fraction $\frac{41}{999.....9(\text{n times})}$. Find n.

(a) 1 (b) 3

(c) 4 (d) 2

RRB Group-D – 01/10/2018 (Shift-III)

Ans : (d) Let $0.\overline{41} = x$

0.414141 ..... = x

Multiplying by 100 in both sides,

100x = 41.4141 .....

100x = 41 + x

99x = 41

$x = \frac{41}{99}$

Hence, it is clear that the number of digit 9 in denominator is twice. So n = 2.

76. Which of the following is equivalent to $0.\overline{56}$? ('—' represents recurring decimal)

(a) $\frac{56}{100}$ (b) $\frac{56}{1000}$

(c) $\frac{56}{99}$ (d) $\frac{560}{90}$

RRB NTPC 17.01.2017 Shift-1

Ans : (c) $0.\overline{56} = 0 + \frac{56}{99}$

$\boxed{0.\overline{56} = \frac{56}{99}}$

77. Correct expression of $1.4\overline{27}$ = ? (Bar sign indicates to recurring decimal)

(a) $\frac{1427}{1000}$ (b) $\frac{157}{110}$

(c) $\frac{1427}{10000}$ (d) $\frac{157}{111}$

RRB NTPC 17.01.2017 Shift-3

Ans : (b) $1.4\overline{27} = 1 + 4\overline{27}$

$= 1 + \frac{427-4}{990}$

$= 1 + \frac{423}{990} = 1 + \frac{47}{110} = \frac{157}{110}$

78. Correct expression of $0.0\overline{18}$ = ?

(a) $\frac{1}{55}$ (b) $\frac{18}{100}$

(c) $\frac{18}{1000}$ (d) $\frac{1}{66}$

RRB NTPC 11.04.2016 Shift : 2

Ans : (a) $x = .0\overline{18} = \frac{18-0}{990}$

$= \frac{18}{990} = \frac{1}{55}$ .....(From decimal fraction system)

79. Correct expression of $0.0\overline{234}$ = ?

(a) $\frac{13}{555}$ (b) $2\frac{34}{100}$

(c) $\frac{134}{990}$ (d) $\frac{234}{1000}$

RRB NTPC 07.04.2016 Shift : 2

Ans : (a) Rule for converting mixed-recurring decimal fraction into common or vulgar fraction:-

First non-recurring part is subtracted from recurring part, as numerator, then take 9, equal to the number of recurring digits, and add 0, equal to the number of non-recurring digits, as denominator.

From question,

$0.0\overline{234} = \frac{234-0}{9990}$

$= \frac{234}{9990} = \frac{78}{3330} = \frac{13}{555}$

80. What is the correct expression of $2.\overline{56}$ (Bar sign represents to recurring decimal)

(a) $2\frac{560}{90}$ (b) $2\frac{56}{99}$

(c) $2\frac{56}{1000}$ (d) $2\frac{56}{100}$

RRB NTPC 19.01.2017 Shift : 3

Ans : (b) $2.\overline{56}$

$2+\frac{56}{99}=2\frac{56}{99}$ ….(From decimal fraction system)

81. Correct expression of $0.02\overline{36}$ = ?

(a) $\frac{13}{550}$ (b) $\frac{236}{1000}$

(c) $2\frac{36}{1000}$ (d) $\frac{13}{555}$

RRB NTPC 22.04.2016 Shift : 2

Ans : (a) $0.02\overline{36}=\frac{236-2}{9900}$

$=\frac{234}{9900}=\frac{13}{550}$

82. Correct expression of $0.126\overline{36}$ = ?

(a) $\frac{139}{1100}$ (b) $126\frac{36}{1000}$

(c) $\frac{139}{2200}$ (d) $126\frac{36}{10000}$

RRB NTPC 26.04.2016 Shift : 2

Ans : (a) from question,

$0.126\overline{36}=\frac{12636-126}{99000}=\frac{12510}{99000}$

$=\frac{1251}{9900}=\frac{139}{1100}$

Type - 3

83. In decimal system $\frac{73}{8}$ can be written:

(a) 9.50 (b) 9.125

(c) 8.150 (d) 8.125

RRB RPF Constable -19/01/2019 (Shift-II)

Ans. (b) : Given fraction, $\frac{73}{8}$,

$\frac{73}{8}=8\overline{)73}$ (9.125)

72

10

8

20

16

40

40

××

Hence, In decimal system $\frac{73}{8}$ can be written 9.125

84. Express $\frac{44}{5}\%+\frac{4}{5}\%+\frac{0.4}{5}\%$ in the form of decimal number.

(a) 0.0888 (b) 0.0998

(c) 0.0896 (d) 0.0968

RRB JE - 26/05/2019 (Shift-III)

Ans : (d) From given expression,

$\frac{44}{5}\%+\frac{4}{5}\%+\frac{0.4}{5}\% = \frac{48.4}{5}\%$

$=\frac{48.4}{500}=\frac{0.484}{5}=0.0968$

85. Which of these fractions cannot be reduced further?

14/21, 33/43, 18/24, 41/82

(a) 33/43 (b) 92/24

(c) 18/24 (d) 41/82

RRB JE - 26/05/2019 (Shift-III)

Ans : (a)

$\frac{14}{21}=\frac{2}{3}$, $\frac{33}{43}=\frac{33}{43}$

$\frac{18}{24}=\frac{3}{4}$

$\frac{41}{82}=\frac{1}{2}$

Hence, it is clear that $\frac{33}{43}$ cannot be reduced further.

86. Reduce to lowest terms 368/575 :–

(a) 28/29 (b) 30/25

(c) 25/29 (d) 16/25

RRB JE - 27/05/2019 (Shift-I)

Ans : (d) Given fraction,

$\frac{368}{575}=\frac{16}{25}$

87. Which of the following fractions equals to 18/25?

(a) 72/100 (b) 36/75

(c) 54/100 (d) 50/100

RRB JE - 30/05/2019 (Shift-III)

Ans : (a)

From the given fraction, $\frac{18}{25}$,

$\Rightarrow \frac{18}{25}=\frac{18\times4}{25\times4}=\frac{72}{100}$

88. Simplify : $\frac{6}{27}\div\frac{27}{30}\div\frac{20}{81}$

(a) 9 (b) 6

(c) 3 (d) 1

RRB RPF Constable -20/01/2019 (Shift-II)

Ans : (d) $\frac{6}{27}\div\frac{27}{30}\div\frac{20}{81}$

$=\frac{6}{27}\times\frac{30}{27}\times\frac{81}{20}=\frac{6\times3\times3}{27\times2}=\frac{6}{3\times2}=1$

89. $\frac{0.3}{1000}$ **equals to :**

(a) 3×10^{-4} (b) 3×10^{-6}
(c) 3×10^{5} (d) 3×10^{-5}

RRB Group-D – 23/09/2018 (Shift-I)

Ans : (a) $\frac{0.3}{1000} = ?$

$\frac{0.3}{1000}$

$= \frac{0.3}{10^3}$

$= 0.3 \times 10^{-3}$

$= 3 \times 10^{-1} \times 10^{-3}$

$? = 3 \times 10^{-4}$

90. Convert $\frac{\frac{4}{9}}{12}$ **in its simple form:**

(a) $\frac{1}{26}$ (b) $\frac{1}{29}$
(c) $\frac{1}{25}$ (d) $\frac{1}{27}$

RRB Group-D – 26/09/2018 (Shift-I)

Ans : (d)

$\frac{\frac{4}{9}}{12} = \frac{4}{9} \times \frac{1}{12} = \frac{1}{27}$

91. $1\frac{2}{3}$ **inversely proportional to :**

(a) $2\frac{2}{3}$ (b) $\frac{3}{5}$
(c) $3\frac{1}{2}$ (d) $\frac{2}{3}$

RRB Group-D – 30/10/2018 (Shift-II)

Ans : (b) $1\frac{2}{3}$

$= \frac{5}{3}$

Writing in inverse $= \frac{3}{5}$

92. The lowest fractional value of 4.025 =?

(a) $\frac{161}{40}$ (b) $\frac{116}{20}$
(c) $\frac{161}{20}$ (d) $\frac{116}{40}$

RRB Group-D – 30/10/2018 (Shift-II)

Ans : (a) The fractional form of 4.025,

$= \frac{4025}{1000}$

$= \frac{161}{40}$

93. Which of the following fractions is not equivalent to 4/11?

(a) $\frac{64}{176}$ (b) $\frac{20}{55}$
(c) $\frac{84}{209}$ (d) $\frac{32}{88}$

RRB Group-D – 27/09/2018 (Shift-I)

Ans. (c) From question,

(a) $\frac{64}{176} = \frac{4}{11}$ (b) $\frac{20}{55} = \frac{4}{11}$

(c) $\frac{84}{209} = 0.401$ (d) $\frac{32}{88} = \frac{4}{11}$

Hence, option (c) is not equivalent to 4/11.

94. $2\frac{1}{25} = ?$

(a) 0.24 (b) 2.4
(c) 2.004 (d) 2.04

RRB Group-D – 27/09/2018 (Shift-III)

Ans : (d) $2\frac{1}{25} = \frac{51}{25} = 2.04$

95. What fraction of a day equals to 7 minutes and 12 second?

(a) $\frac{1}{240}$ (b) $\frac{1}{225}$
(c) $\frac{1}{200}$ (d) $\frac{1}{300}$

RRB Group-D – 16/10/2018 (Shift-II)

Ans : (c) The number of hours in a day = 24
= 24 × 60 × 60 sec

7 min 12 sec = (7 × 60 + 12)
= (420 + 12)
= 432 sec

The required fraction $= \frac{432}{24 \times 60 \times 60} = \boxed{\frac{1}{200}}$

96. Solve : 0.05 × 0.4

(a) 2 (b) 0.2
(c) 0.02 (d) 0.002

RRB NTPC 28.03.2016 Shift : 2

Ans :(c) 0.05 × 0.4 = 0.02

97. The value of 0.000825 ÷ 0.05 = ?

(a) 0.0165 (b) 0.65
(c) 0.00165 (d) 0.015

RRB NTPC 28.03.2016 Shift : 3

Ans : (a) $\frac{0.000825}{0.05} = 0.0165$

98. Find the solution of 0.275 + 0.569 – 0.336 =?

(a) 0.123 (b) 0.508
(c) 0.457 (d) 0.594

RRB NTPC 18.01.2017 Shift : 3

Ans : (b) 0.275 + 0.569 – 0.336
= 0.844–0.336 = 0.508

99. Change $\frac{13}{55}$ in decimal.

(a) 0.20 (b) 0.236
(c) 0.245 (d) 0.257

RRB NTPC 28.04.2016 Shift : 1

Ans :(b) $\frac{13}{55} = 0.236$

100. 414÷54 is written for which mixed fraction of the following?

(a) $7\frac{36}{54}$ (b) $7\frac{6}{9}$
(c) $7\frac{2}{3}$ (d) $7\frac{1}{3}$

RRB NTPC 04.04.2016 Shift : 2

Ans : (c) $414 \div 54 = \frac{414}{54} = \frac{23\times18}{3\times18} = 7\frac{2}{3}$

101. Solve: $\left(\frac{2}{7}+\frac{3}{5}\right)\div\left(\frac{2}{5}+\frac{2}{7}\right)$

(a) $\frac{31}{24}$ (b) $\frac{24}{31}$
(c) $\frac{26}{25}$ (d) $\frac{12}{13}$

RRB NTPC 02.04.2016 Shift : 2

Ans : (a) Given,

$\left(\frac{2}{7}+\frac{3}{5}\right)\div\left(\frac{2}{5}+\frac{2}{7}\right)$

$\left(\frac{10+21}{35}\right)\div\left(\frac{14+10}{35}\right)$

$\frac{31}{35}\div\frac{24}{35} = \frac{31}{35}\times\frac{35}{24} = \frac{31}{24}$

102. Simplify : $\frac{3}{7\frac{1}{3}}+\frac{3}{3\frac{1}{7}}$

(a) $1\frac{3}{11}$ (b) $1\frac{4}{11}$
(c) $2\frac{3}{7}$ (d) $2\frac{4}{7}$

RRB NTPC 29.03.2016 Shift : 2

Ans : (b) $\frac{3}{7\frac{1}{3}}+\frac{3}{3\frac{1}{7}} = \frac{3}{\frac{22}{3}}+\frac{3}{\frac{22}{7}}$

$= \frac{9}{22}+\frac{21}{22}$

$= \frac{30}{22}$

$= \frac{15}{11} = 1\frac{4}{11}$

103. Simplify: $8\frac{1}{3}\times4\frac{1}{5}\div5\frac{1}{4}$

(a) $4\frac{2}{5}$ (b) $5\frac{3}{4}$
(c) $7\frac{1}{3}$ (d) $6\frac{2}{3}$

RRB NTPC 29.03.2016 Shift : 2

Ans : (d) $8\frac{1}{3}\times4\frac{1}{5}\div5\frac{1}{4} = \frac{25}{3}\times\frac{21}{5}\div\frac{21}{4}$

$= \frac{25}{3}\times\frac{21}{5}\times\frac{4}{21}$

$= \frac{20}{3} = 6\frac{2}{3}$

104. Choose the different fraction.
15/20, 48/60, 21/28, 75/100

(a) 15/20 (b) 48/60
(c) 21/28 (d) 75/100

RRB NTPC 28.03.2016 Shift : 3

Ans : (b) $\frac{15}{20} = \frac{5\times3}{5\times4}$, $\frac{48}{60} = \frac{12\times4}{12\times5}$

$\frac{21}{28} = \frac{7\times3}{7\times4}$, $\frac{75}{100} = \frac{25\times3}{25\times4}$

All the fractions are the equivalent fractions of $\frac{3}{4}$, except $\frac{48}{60}$.

105. Find the solution of :– 4/11 + 2/7 + 3/5

(a) 37/35 (b) 481/385
(c) 13/35 (d) 37/385

RRB NTPC 18.01.2017 Shift : 3

Ans : (b) $\frac{4}{11}+\frac{2}{7}+\frac{3}{5}$

$= \frac{35\times4+2\times55+3\times77}{385}$

$= \frac{140+110+231}{385}$

$= \boxed{\frac{481}{385}}$

106. Simplify : $\frac{5}{28}\div\frac{28}{35}\div\frac{20}{112}$

(a) 4/5 (b) 5/4
(c) 4/7 (d) 7/4

RRB NTPC 11.04.2016 Shift : 2

Ans : (b) $\frac{5}{28}\div\frac{28}{35}\div\frac{20}{112}$

$\frac{5}{28}\times\frac{35}{28}\times\frac{112}{20}$

$= \frac{5}{4}$

107. Simplify: $\frac{4}{28} \div \frac{28}{35} \div \frac{20}{112}$

(a) 7 (b) 4
(c) 2 (d) 1

RRB NTPC 07.04.2016 Shift : 2

Ans : (d) Given−

$= \frac{4}{28} \div \frac{28}{35} \div \frac{20}{112}$

$= \frac{4}{28} \times \frac{35}{28} \times \frac{112}{20} = 1$

108. Which of the below given fractions is NOT equal to $\frac{9}{17}$?

(a) $\frac{108}{221}$ (b) $\frac{27}{51}$
(c) $\frac{63}{119}$ (d) $\frac{153}{289}$

RRB ALP & Tec. (31-08-18 Shift-I)

Ans : (a) Given: $\frac{9}{17}$

$\frac{9 \times 3}{17 \times 3} = \frac{27}{51}$

$\frac{9 \times 7}{17 \times 7} = \frac{63}{119}$

$\frac{9 \times 17}{17 \times 17} = \frac{153}{289}$

$\frac{108}{221}$ will not give any equivalent fraction of $\frac{9}{17}$.

109. $\frac{182}{130}$ when written in the simplest form is :

(a) $\frac{28}{20}$ (b) $\frac{91}{65}$
(c) $\frac{14}{10}$ (d) $\frac{7}{5}$

RRB ALP & Tec. (20-08-18 Shift-III)

Ans : (d) $\frac{182}{130}$

$= \frac{91}{65} = \frac{7}{5}$

110. How many kilometres are there in one metre?

(a) 0.0001 (b) 0.1
(c) 0.001 (d) 0.01

RRB ALP & Tec. (10-08-18 Shift-III)

Ans : (c) 1000m = 1km.

$1\text{m} = \frac{1}{1000}\text{km.} = 0.001 \text{ km.}$

Hence, one metre has 0.001 km.

Type - 3

111. The sum of $\frac{5}{12}$ and $\frac{12}{5}$ =?

(a) $\frac{17}{17}$ (b) $\frac{17}{60}$
(c) $\frac{60}{17}$ (d) $\frac{169}{60}$

RRB RPF Constable -17/01/2019 (Shift-III)

Ans. (d) The sum of $\frac{5}{12}$ and $\frac{12}{5} = \frac{5}{12} + \frac{12}{5}$

$= \frac{25+144}{60} = \frac{169}{60}$

112. A fraction added to its inverse is $2\frac{25}{66}$. Find the greater number of the two:

(a) $1\frac{15}{22}$ (b) $1\frac{5}{6}$
(c) $1\frac{20}{33}$ (d) $1\frac{5}{11}$

RRB Group-D – 05/10/2018 (Shift-II)

Ans : (b) Let the fraction be x and its inverse be $\frac{1}{x}$.

According to the question,

$x + \frac{1}{x} = 2\frac{25}{66}$ -----(I)

From option (b),

Putting the value $x = 1\frac{5}{6} = \frac{11}{6}$ in equation (I),

$\frac{11}{6} + \frac{6}{11} = 2\frac{25}{66}$

$\Rightarrow \frac{121+36}{66} = 2\frac{25}{66}$

$\Rightarrow \frac{157}{66} = 2\frac{25}{66}$

$\Rightarrow 2\frac{25}{66} = 2\frac{25}{66}$

Hence greatest fraction $= 1\frac{5}{6}$

113. Sum of $\frac{5}{11}$ and $\frac{11}{5}$ =?

(a) $\frac{146}{55}$ (b) $\frac{16}{16}$
(c) $\frac{16}{55}$ (d) $\frac{110}{55}$

RRB Group-D – 26/09/2018 (Shift-II)

Ans. (a) : $\frac{5}{11} + \frac{11}{5}$

$= \frac{25+121}{55} = \frac{146}{55}$

114. The difference of a positive fraction and its inverse is $6\frac{39}{160}$. Find the fraction.

(a) $\frac{32}{5}$ (b) $\frac{13}{8}$

(c) $\frac{15}{8}$ (d) $\frac{16}{5}$

RRB Group-D – 15/10/2018 (Shift-II)

Ans : (a) Let the positive fraction be x

So, inverse $=\frac{1}{x}$

According to the question,

$$x-\frac{1}{x}=6\frac{39}{160}$$

$$\frac{x^2-1}{x}=\frac{999}{160}$$

$160x^2 - 160 = 999x$

$160x^2 - 999x - 160 = 0$

$160x^2 - (1024–25)x - 160 = 0$

$160x^2 - 1024x + 25x - 160 = 0$

$32x (5x - 32) + 5 (5x - 32) = 0$

$(32x + 5) (5x - 32) = 0$

$32x + 5 = 0$, $5x - 32 = 0$

$32x = - 5$, $5x = 32$

$\boxed{x=\frac{-5}{32}}$, $\boxed{x=\frac{32}{5}}$

115. The difference of a fraction and its inverse is $\frac{9}{11}$. Then the difference of cubes of the fraction and its inverse will be:

(a) $-\frac{1331}{2538}$ (b) $-\frac{2538}{1331}$

(c) $\frac{3996}{1331}$ (d) $\frac{729}{1331}$

RRB Group-D – 11/10/2018 (Shift-I)

Ans : (c)

Let the fraction be $\frac{x}{1}$, then its inverse will be $\frac{1}{x}$,

According to the question,

$$\frac{x}{1}-\frac{1}{x}=\frac{9}{11}$$

$$\Rightarrow x-\frac{1}{x}=\frac{9}{11}$$

Cubing in both side,

$$x^3-\frac{1}{x^3}=\left(\frac{9}{11}\right)^3+3\times\frac{9}{11} \quad \left[a^3-b^3=(a-b)^3+3ab(a-b)\right]$$

$$=\frac{729}{1331}+\frac{27}{11}$$

$$=\frac{729+(27\times121)}{1331}=\frac{729+3267}{1331}$$

$$\therefore x^3-\frac{1}{x^3}=\frac{3996}{1331}$$

116. The sum of a fraction and its inverse is $5\frac{1}{5}$. Find the fraction.

(a) 1/5 (b) 1/6

(c) 1/3 (d) 4

RRB NTPC 29.04.2016 Shift : 1

Ans : (a) Let the fraction be x, so its inverse be $\frac{1}{x}$.

According to the question,

$$x+\frac{1}{x}=5\frac{1}{5}$$

$$\Rightarrow \frac{x^2+1}{x}=\frac{26}{5}$$

$$\Rightarrow 5x^2+5=26x$$

$$\Rightarrow 5x^2-26x+5=0$$

$$\Rightarrow 5x^2-25x-x+5=0$$

$$\Rightarrow 5x(x-5)-1(x-5)=0$$

$$\Rightarrow (5x-1)(x-5)=0$$

$$\therefore 5x-1=0$$

$$\Rightarrow 5x=1$$

$$\Rightarrow x=\frac{1}{5}$$

117. How much does one need to add to $\frac{4}{5}$ to obtain $\frac{5}{4}$?

(a) $\frac{1}{-1}$ (b) $\frac{16}{20}$

(c) $\frac{9}{20}$ (d) $\frac{1.25}{0.8}$

RRB ALP & Tec. (21-08-18 Shift-II)

Ans : (c) Let the adding number be x.

According to the question,

$$x+\frac{4}{5}=\frac{5}{4}$$

$$x=\frac{5}{4}-\frac{4}{5},\ x=\frac{25-16}{20},\ x=\frac{9}{20}$$

Hence $\frac{9}{20}$ is the required number.

118. The sum of $\frac{5}{8}$ and $\frac{8}{5}$ is :

(a) $\frac{13}{13}$ (b) $\frac{80}{40}$

(c) $\frac{13}{40}$ (d) $\frac{89}{40}$

RRB ALP & Tec. (20-08-18 Shift-II)

Ans : (d) According to the question,

Adding the given fractions,

$$\frac{5}{8}+\frac{8}{5}=\frac{25+64}{40}=\frac{89}{40}$$

Type - 3

119. The fraction which, when subtracted from $\frac{1}{2}$ gives $\frac{3}{4}$ is:

(a) $\frac{1}{4}$ (b) $-\frac{1}{4}$

(c) $\frac{1}{3}$ (d) $-\frac{1}{3}$

RRB RPF-SI -12/01/2019 (Shift-III)

Ans. (b) : Let the fraction be $\frac{1}{x}$,

According to the question,

$$\frac{1}{2}-\frac{1}{x}=\frac{3}{4}\Rightarrow-\frac{1}{x}=\frac{3}{4}-\frac{1}{2}\Rightarrow -\frac{1}{x}=\frac{3-2}{4}$$

$$\boxed{\frac{1}{x}=-\frac{1}{4}}$$

Hence, the required fraction is $-\frac{1}{4}$.

120. The sum of two fractions is $\frac{9}{10}$. If one is $\frac{8}{9}$, find the another.

(a) $\frac{1}{5}$ (b) $\frac{1}{1}$

(c) $\frac{1}{90}$ (d) $\frac{72}{90}$

RRB RPF Constable -24/01/2019 (Shift-I)

Ans. (c) : Let the another fraction be $\frac{x}{y}$,

According to the question,

$$\therefore \frac{x}{y}+\frac{8}{9}=\frac{9}{10}$$

$$\frac{x}{y}=\frac{9}{10}-\frac{8}{9}$$

$$\frac{x}{y}=\frac{81-80}{90}$$

$$\frac{x}{y}=\frac{1}{90}$$

Hence, the another fraction is $\frac{1}{90}$.

121. The value of $\frac{3}{15}+\frac{13}{14}-\frac{19}{21}+\frac{31}{35}-\frac{23}{30}$ =?

(a) $\frac{8}{21}$ (b) $\frac{1}{3}$

(c) $\frac{2}{5}$ (d) $\frac{12}{35}$

RRB Group-D – 20/09/2018 (Shift-II)

Ans : (d)

$$\frac{3}{15}+\frac{13}{14}-\frac{19}{21}+\frac{31}{35}-\frac{23}{30}$$

(LCM of 15, 14, 21, 35 and 30 is 210)

$$=\frac{42+195-190+186-161}{210}$$

$$\Rightarrow\frac{423-351}{210}$$

$$\Rightarrow\frac{72}{210}=\frac{12}{35}$$

Hence, the required value is $\frac{12}{35}$.

122. The subtracted value of a fraction from $\frac{1}{6}$ is $\frac{1}{13}$. Find the fraction.

(a) $\frac{7}{78}$ (b) $\frac{5}{13}$

(c) $\frac{1}{7}$ (d) $\frac{11}{39}$

RRB Group-D – 03/10/2018 (Shift-III)

Ans : (a) Let the fraction be $\frac{x}{y}$,

$$=\frac{1}{6}-\frac{x}{y}=\frac{1}{13}$$

$$\frac{x}{y}=\frac{1}{6}-\frac{1}{13}$$

$$\frac{x}{y}=\frac{13-6}{78}$$

$$\frac{x}{y}=\frac{7}{78}$$

Hence, the required fraction is $\frac{7}{78}$.

123. The sum of two fractions is $\frac{7}{4}$. If one is $\frac{5}{3}$, find the another.

(a) $\frac{1}{5}$ (b) $\frac{2}{1}$

(c) $\frac{1}{12}$ (d) $\frac{1}{10}$

RRB Group-D – 18/09/2018 (Shift-II)

Ans. (c) : Let the required fraction be $\frac{x}{y}$,

Then the another fraction is given = $\frac{5}{3}$,

According to the question,

$$\frac{x}{y}+\frac{5}{3}=\frac{7}{4}$$

$$\frac{x}{y}=\frac{7}{4}-\frac{5}{3}=\frac{21-20}{12}=\frac{1}{12}$$

Hence, the required fraction is $\frac{1}{12}$.

124. Find the difference between 0.02 and 0.002.

(a) 0.018 (b) 0.0018
(c) 1.8 (d) 0.18

RRB Group-D – 17/09/2018 (Shift-III)

Ans. (a) : The difference of 0.02 and 0.002,

$\Rightarrow \frac{0.02\times100}{100}-\frac{0.002\times1000}{1000}$

$\Rightarrow \frac{2}{100}-\frac{2}{1000}$

$\Rightarrow \frac{20-2}{1000}$

$\Rightarrow \frac{18}{1000}=0.018$

Hence, the required difference is 0.018.

125. Which fraction of the following when subtracted from $\frac{3}{4}$ is $\frac{5}{12}$?

(a) $\frac{1}{3}$ (b) $\frac{2}{8}$
(c) $\frac{1}{6}$ (d) $\frac{2}{3}$

RRB Group-D – 19/09/2018 (Shift-III)

Ans. (a) : Let the fraction be $\frac{1}{x}$,

According to the question,

$\frac{3}{4}-\frac{1}{x}=\frac{5}{12}$

$-\frac{1}{x}=\frac{5}{12}-\frac{3}{4}$

$-\frac{1}{x}=\frac{20-36}{48}$

$-\frac{1}{x}=\frac{-16}{48}$

$\frac{1}{x}=\frac{1}{3}$

Hence, the required fraction is $\frac{1}{3}$.

126. What should be added to $\frac{10}{11}$ to get $\frac{11}{10}$?

(a) $\frac{21}{110}$ (b) $\frac{1}{-1}$
(c) $\frac{1}{55}$ (d) $\frac{2}{11}$

RRB Group-D – 20/09/2018 (Shift-I)

Ans. (a) : Let the adding number be x.

According to the question,

$\frac{10}{11}+x=\frac{11}{10}$

$x=\frac{11}{10}-\frac{10}{11}=\frac{121-100}{110}=\frac{21}{110}$

Hence the adding number is $\frac{21}{110}$.

127. What should be added to $5\frac{3}{5}$ to get $8\frac{3}{7}$?

(a) $\frac{99}{35}$ (b) $\frac{96}{35}$
(c) $\frac{99}{33}$ (d) $\frac{94}{35}$

RRB Group-D – 25/09/2018 (Shift-III)

Ans. (a) : Let the required number be x.

According to the question,

$5\frac{3}{5}+x=8\frac{3}{7}$

$x = 8\frac{3}{7}-5\frac{3}{5} = \frac{59}{7}-\frac{28}{5}$

$=\frac{295-196}{35} = \frac{99}{35}$

Hence, the required number is $\frac{99}{35}$

128. The sum of two fractions is $\frac{7}{6}$. One is $\frac{3}{4}$. Find the other.

(a) $\frac{4}{12}$ (b) $\frac{5}{12}$
(c) $\frac{4}{2}$ (d) $\frac{1}{12}$

RRB Group-D – 26/09/2018 (Shift-III)

Ans : (b) Let the other fraction is x.

According to the question,

$\Rightarrow x+\frac{3}{4}=\frac{7}{6} \Rightarrow x=\frac{7}{6}-\frac{3}{4}$

$\Rightarrow x=\frac{14-9}{12}=\frac{5}{12}$

Hence, the required fraction is $\frac{5}{12}$.

129. A fraction when added to $\frac{7}{3}$, gives 4. Find the fraction.

(a) $1\frac{2}{3}$ (b) $\frac{11}{2}$
(c) $-\frac{1}{2}$ (d) $\frac{2}{3}$

RRB Group-D – 28/09/2018 (Shift-I)

Ans : (a) Let the required fraction be x.

According to the question,

$\Rightarrow \frac{x}{1}+\frac{7}{3}=4$

$\Rightarrow \frac{3x+7}{3}=4$

$\Rightarrow 3x+7=4\times3$

$\Rightarrow 3x+7=12 \Rightarrow 3x=12-7 \Rightarrow 3x=5$

$\Rightarrow x=\frac{5}{3}=\left(1\frac{2}{3}\right)$

Hence, the required fraction is $1\frac{2}{3}$.

130. The difference between two fractions is $\frac{5}{6}$. The smaller one is $\frac{3}{4}$. Find the other.

(a) $\frac{1}{12}$ (b) $\frac{19}{24}$

(c) $\frac{19}{12}$ (d) $\frac{8}{10}$

RRB Group-D – 22/10/2018 (Shift-II)

Ans : (c) Let the other fraction be x.
According to the question,

$$x-\frac{3}{4}=\frac{5}{6}$$

$$\Rightarrow x=\frac{5}{6}+\frac{3}{4}$$

$$x=\frac{38}{24}=\frac{19}{12}$$

Hence, the other fraction is $\frac{19}{12}$.

131. The value of $\frac{5}{3}+\frac{3}{5}$=?

(a) $\frac{15}{8}$ (b) $\frac{8}{15}$

(c) $2\frac{4}{15}$ (d) $\frac{8}{8}$

RRB Group-D – 22/10/2018 (Shift-II)

Ans : (c) From given expression,

$$\Rightarrow \frac{5}{3}+\frac{3}{5}$$

$$=\frac{25+9}{15}$$

$$=\frac{34}{15}=2\frac{4}{15}$$

132. The difference of $\frac{25}{12}$ and $\frac{15}{8}$ = ?

(a) $\frac{10}{24}$ (b) $\frac{7}{13}$

(c) $\frac{10}{4}$ (d) $\frac{5}{24}$

RRB Group-D – 06/12/2018 (Shift-II)

Ans. (d) The difference of $\frac{25}{12}$ and $\frac{15}{8}$,

$$\frac{25}{12}-\frac{15}{8}=\frac{50-45}{24}=\frac{5}{24}$$

133. To get $\frac{5}{12}$, $\frac{25}{3}$ should be multiplied by:

(a) 10 (b) 20

(c) $\frac{4}{5}$ (d) $\frac{5}{4}$

RRB Group-D – 05/12/2018 (Shift-II)

Ans. (b) Let the required number be x.
According to the question,

$$\frac{5}{12}\times x=\frac{25}{3}$$

$$\frac{x}{4}=5$$

$$x=20$$

134. The square root of a positive fraction, when added to 1, is $3\frac{1}{4}$. Find the fraction.

(a) $2\frac{1}{4}$ (b) $6\frac{1}{4}$

(c) $5\frac{1}{16}$ (d) $3\frac{1}{16}$

RRB Group-D – 02/11/2018 (Shift-I)

Ans. (c)

Let the fraction be $=\frac{x}{y}$

According to the question,

$$\sqrt{\frac{x}{y}}+1=3\frac{1}{4}$$

$$\sqrt{\frac{x}{y}}=\frac{13}{4}-1$$

$$\sqrt{\frac{x}{y}}=\frac{9}{4}$$

$$\frac{x}{y}=\frac{81}{16},\quad \frac{x}{y}=5\frac{1}{16}$$

135. The difference between $\frac{11}{12}$ and $\frac{7}{8}$=?

(a) $\frac{1}{4}$ (b) $\frac{4}{4}$

(c) $\frac{4}{24}$ (d) $\frac{1}{24}$

RRB Group-D – 11/12/2018 (Shift-III)

Ans : (d) The difference between $\frac{11}{12}$ and $\frac{7}{8}$

$$=\frac{11}{12}-\frac{7}{8}$$

$$=\frac{88-84}{96}$$

$$=\frac{4}{96}$$

$$=\frac{1}{24}$$

Hence, the required difference is $\frac{1}{24}$.

136. What is the fraction which, when taken away from $\frac{3}{4}$, gives $\frac{2}{5}$?

(a) $-\frac{1}{1}$ (b) $\frac{7}{20}$

(c) $\frac{1}{20}$ (d) $\frac{3}{10}$

RRB Group-D – 08/10/2018 (Shift-III)

Ans : (b) Let the fraction be $\frac{x}{y}$.

According to the question,

$$\frac{3}{4}-\frac{x}{y}=\frac{2}{5}$$

$$\Rightarrow \frac{x}{y}=\frac{3}{4}-\frac{2}{5}$$

$$\Rightarrow \frac{x}{y}=\frac{15-8}{20}$$

$$\frac{x}{y}=\frac{7}{20}$$

137. What should be added to 3/5 to get 5/4?

(a) $\frac{15}{20}$ (b) $\frac{13}{20}$

(c) $\frac{2}{-1}$ (d) $\frac{1.25}{0.6}$

RRB Group-D – 05/10/2018 (Shift-III)

Ans. (b) Let the required number be x.

According to the question,

$$\frac{5}{4}=\frac{3}{5}+x$$

$$\frac{5}{4}-\frac{3}{5}=x$$

$$\frac{25-12}{20}=x$$

$$x=\frac{13}{20}$$

Hence, the required number is $\frac{13}{20}$.

138. A fraction is subtracted from 1/5 and gives 1/12. Find the fraction.

(a) $\frac{1}{7}$ (b) $\frac{11}{30}$

(c) $\frac{7}{60}$ (d) $\frac{5}{12}$

RRB Group-D – 04/10/2018 (Shift-I)

Ans. (c) Let the fraction be $\frac{x}{y}$.

According to the question,

$$\frac{1}{5}-\frac{x}{y}=\frac{1}{12}$$

$$\frac{1}{5}-\frac{1}{12}=\frac{x}{y}$$

$$\frac{12-5}{60}=\frac{x}{y}$$

$$\frac{x}{y}=\frac{7}{60}$$

139. A fraction when added to $\frac{17}{3}$ is 4. Find the fraction.

(a) $\frac{2}{3}$ (b) $-\frac{1}{1}$

(c) $-1\frac{2}{3}$ (d) $\frac{9}{2}$

RRB Group-D – 01/10/2018 (Shift-III)

Ans : (c) Let the required fraction be $\frac{x}{y}$.

According to the question,

$$\therefore \frac{x}{y}+\frac{17}{3}=4$$

$$\frac{x}{y}=4-\frac{17}{3}$$

$$=\frac{12-17}{3}=\frac{-5}{3}$$

$$\frac{x}{y}=-1\frac{2}{3}$$

140. In which fraction, when 5/16 is added gives 1?

(a) $\frac{11}{32}$ (b) $\frac{13}{2}$

(c) $\frac{22}{32}$ (d) $\frac{6}{8}$

RRB Group-D – 19/09/2018 (Shift-I)

Ans : (c) Let the fraction be x.

$$x+\frac{5}{16}=1, \qquad x=1-\frac{5}{16}$$

$$x=\frac{11}{16}, \qquad x=\frac{2\times 11}{2\times 16}=\frac{22}{32}$$

Hence, the require fraction is $\frac{22}{32}$.

141. What is the fraction which, when taken away from $\frac{1}{2}$, gives $\frac{2}{3}$?

(a) $\frac{1}{3}$ (b) $-\frac{1}{3}$

(c) $-\frac{1}{6}$ (d) $\frac{1}{6}$

RRB ALP & Tec. (20-08-18 Shift-I)

Ans : (c) Let the fraction be $\frac{x}{y}$.

According to the problem,

$$\frac{1}{2}-\frac{x}{y}=\frac{2}{3} \qquad \frac{x}{y}=\frac{1}{2}-\frac{2}{3}$$

$$\frac{x}{y}=\frac{-1}{6}$$

142. The sum of two fractions is $\frac{3}{4}$ One of them is $\frac{2}{3}$. What is the other fraction?

(a) $\frac{1}{5}$ (b) $\frac{1}{12}$

(c) $\frac{1}{10}$ (d) $\frac{1}{1}$

RRB ALP & Tec. (20-08-18 Shift-II)

Ans : (b) The sum of fractions = $\frac{3}{4}$

The first fraction = $\frac{2}{3}$

$\therefore$ The other fraction = $\frac{3}{4}-\frac{2}{3}$

$$=\frac{9-8}{12}=\frac{1}{12}$$

143. How much does one need to add to $\frac{2}{3}$ to obtain $\frac{3}{2}$?

(a) $\frac{4}{9}$ (b) $\frac{5}{6}$

(c) $\frac{1}{-1}$ (d) $\frac{1.5}{6}$

RRB ALP & Tec. (17-08-18 Shift-I)

Ans : (b) Let the adding number is x.

According to the question,

$$\frac{2}{3}+x=\frac{3}{2}$$

$$x=\frac{3}{2}-\frac{2}{3}=\frac{9-4}{6}=\frac{5}{6}$$

144. A fraction, when subtracted from $\frac{1}{3}$ gives $\frac{1}{12}$. The fraction is:

(a) $\frac{5}{12}$ (b) $\frac{1}{4}$

(c) $\frac{3}{4}$ (d) $\frac{1}{9}$

RRB ALP & Tec. (14-08-18 Shift-I)

Ans : (b) Let the fraction be $\frac{x}{y}$.

According to the question,

$$\frac{1}{3}-\frac{x}{y}=\frac{1}{12}$$

$$\frac{x}{y}=\frac{1}{3}-\frac{1}{12}=\frac{4-1}{12}=\frac{3}{12}=\frac{1}{4}$$

$$\frac{x}{y}=\frac{1}{4}$$

145. The sum of two fractions is $\frac{5}{6}$. One of them is $\frac{3}{4}$. What is the other fraction ?

(a) $\frac{2}{5}$ (b) $\frac{1}{10}$

(c) 1 (d) $\frac{1}{12}$

RRB ALP & Tec. (10-08-18 Shift-I)

Ans : (d) Let the other fraction be x.

According to the question,

$$x+\frac{3}{4}=\frac{5}{6}$$

$$\therefore \quad x=\frac{5}{6}-\frac{3}{4}=\frac{10-9}{12}=\frac{1}{12}$$

146. Which of the fractions given below, when added to $\frac{5}{7}$ gives 1 ?

(a) $\frac{4}{2}$ (b) $\frac{6}{21}$

(c) $\frac{5}{3}$ (d) $\frac{6}{14}$

RRB ALP & Tec. (30-08-18 Shift-II)

Ans : (b) Let the adding fraction be x.

According to the question,

$$x=1-\frac{5}{7}=\frac{2}{7}$$

Multiplying by 3 in the numerator and denominator of $\frac{2}{7}$,

$$=\frac{2\times3}{7\times3}=\frac{6}{21}$$

Hence, the required fraction is option (b) = $\frac{6}{21}$.

147. Which of the fractions given below, when added to $\frac{5}{8}$, gives 1?

(a) $\frac{6}{24}$ (b) $\frac{5}{2}$

(c) $\frac{6}{16}$ (d) $\frac{6}{3}$

RRB ALP & Tec. (09-08-18 Shift-II)

Ans : (c) Let the fraction be x.
According to the question,

$$\frac{5}{8}+x=1$$

$$x=1-\frac{5}{8}$$

$$x=\frac{3\times2}{8\times2}$$

$$x=\frac{6}{16}$$

Hence the required fraction is option (c) $=\frac{6}{16}$.

148. Solve the following:
196 – 19.6 – 1.96 – 0.196 = ?
(a) 173.254 (b) 173.234
(c) 174.234 (d) 174.244

RRB ALP & Tec. (31-08-18 Shift-I)

Ans : (d) Given,
196−19.6−1.96−0.196
= 196–21.756
= 174.244

149. 0.196 + 1.96 + 19.6 + 196 = ?
(a) 217.756 (b) 216.856
(c) 216.756 (d) 217.676

RRB ALP & Tec. (20-08-18 Shift-III)

Ans : (a) Given,
0.196 + 1.96 + 19.6 + 196 = 217.756

150. Solve the following :
123 + 12.3 + 1.23 + 0.123 + 0.0123 = ?
(a) 136.7659 (b) 136.653
(c) 136.6653 (d) 136.7760

RRB ALP & Tec. (17-08-18 Shift-II)

Ans : (c) Given,
123+12.3+1.23+0.123+0.0123=136.6653

Type - 8

151. If 0.001 ÷ x = 0.01, then find 'x'.
(a) 0.5 (b) 1
(c) 0.01 (d) 0.1

RRB JE - 23/05/2019 (Shift-II)

Ans : (d) Given,
0.001 ÷ x = 0.01

$$\frac{0.001}{x}=0.01$$

$$\frac{0.1}{x}=1\Rightarrow x=0.1$$

Hence, the value of x is 0.1 that is option (d).

152. The value of 0.0006697 to three digits of decimal will be:
(a) 0.000670 (b) 0.00669
(c) 0.001 (d) 0

RRB RPF Constable -22/01/2019 (Shift-I)

Ans : (c) The value of 0.0006697 till three digits of decimal = 0.001,
After decimal if the right digit is 5 or more than 5, then we add 1 to the left digit.

153. Which fraction is not equal to $\frac{15}{23}$?
(a) $\frac{105}{162}$ (b) $\frac{75}{115}$
(c) $\frac{45}{69}$ (d) $\frac{30}{46}$

RRB RPF-SI -12/01/2019 (Shift-I)

Ans : (a) From question,
(a) $\frac{105}{162}=\frac{35}{54}$
(b) $\frac{75}{115}=\frac{15}{23}$
(c) $\frac{45}{69}=\frac{15}{23}$
(d) $\frac{30}{46}=\frac{15}{23}$

Hence, it is clear that option (a) is not equal to $\frac{15}{23}$.

154. 0.065 × 0.4 = ?
(a) 0.26 (b) 0.026
(c) 2.6 (d) 0.0026

RRB RPF-SI -11/01/2019 (Shift-I)

Ans : (b) Given,
0.065 × 0.4 = 0.026

155. Find the value of 0.1404 ÷ 0.06 = ?
(a) 0.234 (b) 2.34
(c) 234 (d) 23.4

RRB RPF Constable -18/01/2019 (Shift-I)

Ans :(b) Given,
0.1404 ÷ 0.06

$$=\frac{0.1404\times10000}{0.06\times10000}=\frac{1404}{600}=2.34$$

156. Find the sum of the place value of 5 and 4 in $\frac{6}{8}$ and $\frac{6}{25}$ respectively.
(a) $\frac{8}{100}$ (b) $\frac{99}{100}$
(c) $\frac{9}{100}$ (d) $\frac{88}{100}$

RRB Group-D – 28/09/2018 (Shift-III)

Ans : (c) $\frac{6}{8}=0.75$

The place value of 5, in 0.75 = $0.05=\frac{5}{100}$

and, $\frac{6}{25}=0.24$

The place value of 4, in 0.24 = $0.04 = \frac{4}{100}$

So, the required sum of both values

$$= \frac{5}{100} + \frac{4}{100} = \frac{9}{100}$$

157. The whole value of 0.008594 to three digits of decimal will be?

(a) 0.008 (b) 0.009
(c) 0.00860 (d) 0.00859

RRB Group-D – 08/10/2018 (Shift-II)

Ans : (b) As we know :- After decimal if the right digit is 5 or more than 5, then we add 1 to the left digit.
Hence, the whole value of 0.008594 to the three digits of decimal will be 0.009.

158. x and y, given correct to 2 decimal place, are given as 4.51 and 2.48 respectively. What is the upper bound of the value of x + y? (The value is correct to the two digits of decimal.)

(a) 7.000 (b) 6.995
(c) 7.010 (d) 6.990

RRB Group-D – 12/10/2018 (Shift-III)

Ans : (a) x + y = 4.51 + 2.48 = 6.99
Hence, the nearest upper limit for the value of (x + y) that is 6.99 = 7.000.

159. Find the value of x.

$$\frac{144}{0.144} = \frac{14.4}{x}$$

(a) 0.0001 (b) 0.0144
(c) 0.1 (d) 0.01

RRB Group-D – 16/10/2018 (Shift-I)

Ans. (b) : Given expression,

$$\frac{144}{0.144} = \frac{14.4}{x}$$

$$144 \times x = 14.4 \times 0.144$$

$$x = \frac{14.4 \times 0.144}{144}$$

$$x = \frac{144 \times 0.144}{144 \times 10}$$

$$x = 0.0144$$

160. If x is integer 0.80000, then what is interval of x?

(a) $0.79995 < x \le 0.80005$
(b) $0.799905 \le x < 0.800005$
(c) $0.799995 \le x < 0.800005$
(d) $0.79995 \le x < 0.80005$

RRB Group-D – 30/10/2018 (Shift-I)

Ans : (c)
The required interval of x = $0.799995 \le x < 0.800005$

161. In $\frac{0.7}{1-6c}$ = -0.2 , then c = ?

(a) 0.8 (b) 0.5
(c) 0.75 (d) 0.075

RRB Group-D – 20/09/2018 (Shift-I)

Ans. (c) : Given,

$$\frac{0.7}{1-6c} = -0.2$$

$$-0.2 + 1.2c = 0.7$$

$$1.2c = 0.9$$

$$c = \frac{0.9}{1.2}$$

$$c = 0.75$$

162. x is written as 15.84, to two digits of decimal. Which of the following is true?

(a) $15.835 < x \le 15.845$
(b) $15.835 < x < 15.845$
(c) $15.835 \le x \le 15.845$
(d) $15.835 \le x < 15.845$

RRB Group-D – 09/10/2018 (Shift-I)

Ans. (d) : Analyzing option (d),
15.835 is less than x, and 15.835 is written to two digits of decimal as 15.84 (approx). While 15.845 will be definitely greater.

163. If x is 0.70000, to five digits of decimal, then interval of x will be:

(a) $0.6995 \le x < 0.70005$
(b) $0.699995 \le x < 0.700005$
(c) $0.699905 \le x < 0.700005$
(d) $0.69995 < x \le 0.70005$

RRB Group-D – 10/10/2018 (Shift-III)

Ans : (b) If x is 0.70000, to five digits of decimal,
Then, $0.699995 \le x < 0.700005$ is correct.

164. The least value of x which makes $\frac{65}{x-14}$ an integer, is:

(a) 1 (b) –51
(c) 79 (d) –1

RRB Group-D – 26/10/2018 (Shift-III)

Ans : (b) From question,
Putting the value of options in the place of x.

(a) $\frac{65}{1-14} = \frac{65}{-13} = -5$ (Integer)

(b) $\frac{65}{-51-14} = \frac{65}{-65} = -1$ is also an integer for which the value of x is the least.

(c) $\frac{65}{79-14} = \frac{65}{65} = 1$ (Integer)

(d) $\frac{65}{-1-14} = \frac{65}{-15} = -4.33$ (Non-Integer)

Hence, it is clear that the least required value of x is -1

165. The product of $\frac{144}{100}$ and $\frac{175}{216}$ will be = ?

(a) $\frac{7}{12}$ (b) $\frac{14}{3}$
(c) $\frac{7}{6}$ (d) $\frac{7}{3}$

RRB Group 'D' 07/12/2018 (Shift-I)

Ans : (c) $\frac{144}{100} \times \frac{175}{216}$

$= \frac{7}{6}$

166. x and y, are given correct to the two digits of decimal, are written as 3.57 and 3.42 respectively. What is the upper limit for x + y ?

(a) 7.000 (b) 7.010
(c) 6.990 (d) 6.995

RRB Group-D – 23/10/2018 (Shift-I)

Ans. (a) : According to the question,
x and y are correct to the two digits of decimal.
x = 3.57 and y = 3.42,
then, x + y = 3.57 + 3.42 = 6.99
hence, the upper limit of x + y = 7.000

167. x and y, are given correct to the two digits of decimal, are written as 2.51 and 3.50 respectively. What is the lower limit for x + y ?

(a) 6.010 (b) 5.995
(c) 6.000 (d) 5.990

RRB Group-D – 15/10/2018 (Shift-II)

Ans : (c) According to the question,
x = 2.51 and y = 3.50
then,
x + y = 2.51 + 3.50 = $\boxed{6.01}$
So the lower limit for x + y is 6.000

168. Find the value of x.

$$\frac{484}{4.84} = \frac{48.4}{x}$$

(a) 0.484 (b) 0.00484
(c) 0.0484 (d) 4.84

RRB Group-D – 08/10/2018 (Shift-III)

Ans : (a) $\frac{484}{4.84} = \frac{48.4}{x}$

$\Rightarrow \quad x \times 484 = 48.4 \times 4.84$

$\Rightarrow \quad x = \frac{484 \times 484}{484 \times 1000}$

Hence, x = 0.484

169. x and y, are given correct to the one digit of decimal, and written as 6.2 and 1.3 respectively. What is the upper limit of $\frac{x}{y}$?

(a) 4.96 (b) 5
(c) 4.77 (d) 5.05

RRB Group-D – 04/10/2018 (Shift-I)

Ans. (b) $\frac{x}{y} = \frac{6.2}{1.3} = 4.76$

Hence, the value of upper limit of $\frac{x}{y}$ is 5.

170. 1.008 = ?

(a) $1\frac{1}{125}$ (b) $1\frac{3}{25}$
(c) $1\frac{2}{25}$ (d) $1\frac{2}{125}$

RRB Group-D – 01/10/2018 (Shift-II)

Ans. (a) : 1.008 = ?

$\Rightarrow \quad \frac{1008}{1000} = \frac{504}{500} = \frac{252}{250} = \frac{126}{125} = 1\frac{1}{125}$

Hence, the value of 1.008 is $1\frac{1}{125}$.

171. If $X = \frac{63.5535}{13.05}$, find the value of X.

(a) 4.48 (b) 4.87
(c) 4.46 (d) 4.28

RRB Group-D – 23/09/2018 (Shift-II)

Ans : (b) $X = \frac{63.5535}{13.05}$

$X = \frac{6355.35}{1305} = 4.87$

Hence, the value of X is 4.87

172. If $\frac{1}{43.21} = 0.2314$, then $\frac{1}{0.0004321} = ?$

(a) 23.14 (b) 2314.27
(c) 0.0002314 (d) 231.4

RRB NTPC 19.01.2017 Shift : 2

Ans :(b) Given,

$\Rightarrow \frac{1}{43.21} = 0.2314$

$= \frac{1}{43.21 \times 10^{-5}} = \frac{10^5}{43.21} = 2314.27$

173. $\frac{3.24 \times 4}{0.2} = ?$

(a) $\frac{324}{25}$ (b) $\frac{162}{25}$
(c) $\frac{162}{2}$ (d) $\frac{324}{5}$

RRB NTPC 19.01.2017 Shift : 3

Ans :(d) $\frac{3.24 \times 4}{0.2} = \frac{3.24 \times 4 \times 100}{0.2 \times 100}$

$= \frac{324 \times 4}{20} = \boxed{\frac{324}{5}}$

174. If $\sqrt{5} = 2.236$; then $\sqrt{5}/2$ is

(a) 1.851 (b) 1.118
(c) 2.236 (d) 1.782

RRB NTPC 27.04.2016 Shift : 3

Ans :(b)

$\frac{\sqrt{5}}{2} = \frac{2.236}{2} = 1.118$

$(\because \sqrt{5} = 2.236)$

175. Find the value of '?' – If 0.0043/?= 0.43

(a) 0.10 (b) 0.01
(c) 1.001 (d) 0.001

RRB NTPC 12.04.2016 Shift : 2

Ans : (b) $\frac{0.0043}{x} = 0.43$

$x = \frac{0.0043}{0.43}$

$x = \frac{43}{4300} = \frac{1}{100}$

$x = 0.01$

176. What is the value of 0.968 ÷ 0.11 = ?

(a) 88 (b) 8.8
(c) 0.88 (d) 8

RRB NTPC 30.04.2016 Shift : 3

Ans : (b) 0.968 ÷ 0.11

$= \frac{968}{110}$

$= 8.8$

177. If 2334/33.1 = 261, then 23.34/3.31 = ?

(a) 0.261 (b) 2.61
(c) 26.1 (d) 261

RRB NTPC 18.04.2016 Shift : 3

Ans : (c) Given,

$\frac{2334}{33.1} = 261.....(1)$

$\because \frac{23.34}{3.31} = \frac{2334}{331}$

$= \frac{2334}{33.1 \times 10}$

$= \frac{2334}{33.1} \times \frac{1}{10}$

$= \frac{261}{10}$ {from equation (1)}

$= 26.1$

178. 23 × 19 = 437. How much is 0.0437 ÷ 1.9 = ?

(a) 0.0023 (b) 2.3
(c) 0.023 (d) 0.23

RRB ALP & Tec. (21-08-18 Shift-I)

Ans : (c) Given,

$23 \times 19 = 437$

According to the question,

$\frac{0.0437}{1.9} = \frac{0.0437 \times 10000}{1.9 \times 10000}$

$= \frac{23 \times 19}{19 \times 1000}$

$= \frac{23}{1000} = 0.023$

179. 17 x 29 = 493. How much is 1700 x 0.0029 = ?

(a) 0.493 (b) 0.0493
(c) 4.93 (d) 49.3

RRB Group-D – 19/09/2018 (Shift-III)

Ans. (c) : 17×29 = 493, तो 1700×0.0029 = ?

⇒ 1700×0.0029

⇒ 17×100×0.0029

⇒ 17×0.29

⇒ 4.93

180. If 493 ÷ 29 = 17, then 4.93÷ 0.0017 = ?

(a) 290 (b) 0.29
(c) 2.9 (d) 2900

RRB ALP & Tec. (13-08-18 Shift-I)

Ans : (d) Given,

$493 \div 29 = 17$,

So,

$4.93 \div 0.0017 = \frac{4.93}{0.0017}$

$= \frac{4.93 \times 10000}{0.0017 \times 10000} = \frac{49300}{17}$

$= 2900$

181. 23 × 31=713. How much is 0.0713 ÷ 3.1?

(a) 0.0023 (b) 0.23
(c) 0.023 (d) 2.3

RRB ALP & Tec. (13-08-18 Shift-III)

Ans : (c) Given,

$23 \times 31 = 713$

So,

$0.0713 \div 3.1 = \frac{0.0713}{3.1}$

$= \frac{0.0713 \times 10000}{3.1 \times 10000} = \frac{713}{31000}$

$= 0.023$

182. Given 17 × 29 = 493, then 170 × 0.029 = ?

(a) 0.493 (b) 4.93
(c) 0.0493 (d) 49.3

RRB ALP & Tec. (10-08-18 Shift-III)

Ans : (b) Given,

$17 \times 29 = 493$

So,

$170 \times 0.029 = 17 \times 10 \times \frac{29}{1000}$

$= \frac{17 \times 29}{100} = \frac{493}{100} = 4.93$

Type - 9

183. Which of the following fractions is greater than 7/12, and smaller than 11/16?

(a) 1/2 (b) 5/8
(c) 7/8 (d) 3/8

RRB JE - 27/06/2019 (Shift-I)

Ans : (b) From the given fractions,

$\frac{7}{12} = 0.58$ $\frac{11}{16} = 0.68$

From question,

(a) $\frac{1}{2} = 0.50$ (b) $\frac{5}{8} = 0.62$

(c) $\frac{7}{8} = 0.87$ (d) $\frac{3}{8} = 0.37$

Hence, option (b) is 0.62, which is greater than 0.58 and smaller than 0.68.

184. Which of the following is correct?

(a) $\frac{9}{16} \le \frac{13}{24}$ (b) $\frac{9}{16} > \frac{13}{24}$

(c) $\frac{9}{16} = \frac{13}{24}$ (d) $\frac{9}{16} < \frac{13}{24}$

RRB Group-D – 17/09/2018 (Shift-I)

Ans : (b) From options,

(a) $\frac{9}{16} \le \frac{13}{24} = 0.56 \le 0.54$ (wrong)

(b) $\frac{9}{16} > \frac{13}{24} = 0.56 > 0.54$ (right)

(c) $\frac{9}{16} = \frac{13}{24} = 0.56 = 0.54$ (wrong)

(d) $\frac{9}{16} < \frac{13}{24} = 0.56 < 0.54$ (wrong)

185. Which of the following is a commutative fraction?

(a) $\frac{77}{72}$ (b) $\frac{5}{49}$

(c) $\frac{105}{112}$ (d) $\frac{104}{121}$

RRB Group-D – 04/10/2018 (Shift-I)

Ans. (b) From question,

$\Rightarrow \frac{77}{72} = 1.0694$

$\Rightarrow \frac{5}{49} = 0.1020$

$\Rightarrow \frac{105}{112} = 0.9375$

$\Rightarrow \frac{104}{121} = 0.8595$

Hence, the commutative fraction is $\frac{5}{49}$.

186. Find the smallest of the following decimals.

(a) 0.1 × 0.1 × 0.1 (b) 0.03 / 3

(c) 0.01 / 2 (d) 0.1 × 0.02 × 0.2

RRB NTPC 05.04.2016 Shift-1

Ans : (d) From question,

(a) 0.1 × 0.1 × 0.1 = 0.001

(b) 0.03 / 3 = 0.01

(c) 0.01 / 2 = 0.005

(d) 0.1 × 0.02 × 0.2 = 0.0004

187. Find the smallest of the following decimals.

(a) 0.2 × 0.2 × 0.2 (b) 0.02 / 3

(c) 0.01 / 2 (d) 0.1 × 0.02 × 2

RRB NTPC 31.03.2016 Shift : 2

Ans : (d) From question,

$\Rightarrow$ 0.2 × 0.2 × 0.2 = 0.008

$\Rightarrow \frac{0.02}{3} = 0.0067$

$\Rightarrow \frac{0.01}{2} = 0.005$

$\Rightarrow$ 0.1 × 0.02 × 2 = 0.004

Hence, it is clear that option (d) is the smallest.

188. Find the fraction which is as much greater than $\frac{4}{7}$ as it is less than $\frac{5}{6}$:

(a) $\frac{59}{84}$ (b) $\frac{84}{59}$

(c) $\frac{58}{84}$ (d) $\frac{59}{85}$

RRB ALP & Tec. (10-08-18 Shift-I)

Ans : (a) Let the fraction is x.

According to the question,

$$\frac{4}{7} < x > \frac{5}{6}$$

Then,

$$x\left(\text{Middle fraction}\right) = \frac{\text{Sum of both fractions}}{2}$$

$$x = \frac{\frac{4}{7}+\frac{5}{6}}{2} = \frac{\frac{24+35}{42}}{2} = \frac{59}{84}$$

189. The product of two numbers is 0.432. One of the number is 1.6. What is the other number?

(a) 2.7 (b) 0.027

(c) 0.27 (d) 27

RRB ALP & Tec. (10-08-18 Shift-I)

Ans : (c)

$$\text{The required number} = \frac{\text{Prouduct of both numbers}}{\text{First number}}$$

$$= \frac{0.432}{1.6}$$

$$= 0.27$$

190. Which of the following is true?

(a) $\frac{29}{6} = \frac{53}{12}$ (b) $\frac{29}{6} = \frac{43}{12}$

(c) $\frac{29}{6} > \frac{43}{12}$ (d) $\frac{29}{6} < \frac{43}{12}$

RRB ALP & Tec. (09-08-18 Shift-I)

Ans : (c) From question,

(a) $\frac{29}{6} = \frac{53}{12}$ (false)

(b) $\frac{29}{6} = \frac{43}{12}$ (false)

(c) $\frac{29}{6} > \frac{43}{12}$ (true)

(d) $\frac{29}{6} < \frac{43}{12}$ (false)

Type - 8

191. If 3 is added to the numerator and the denominator of a fraction, it becomes $\frac{10}{11}$. And if 4 is subtracted from the numerator and the denominator, then it becomes $\frac{3}{4}$. What is the fraction?

(a) $\frac{7}{8}$ (b) $\frac{6}{13}$

(c) $\frac{3}{4}$ (d) $\frac{3}{5}$

RRB RPF Constable -25/01/2019 (Shift-III)

Ans : (a) Let the fraction is $\frac{x}{y}$

According to the question,

$$\frac{x+3}{y+3}=\frac{10}{11}$$

$$\Rightarrow 11x+33=10y+30$$

$$\Rightarrow 11x-10y=-3.....(i)$$

Again, $\frac{x-4}{y-4}=\frac{3}{4}$

$$\Rightarrow 4x-16=3y-12$$

$$\Rightarrow 4x-3y=4 .....(ii)$$

From equation (i) and (ii)–

x = 7 and y = 8

Hence, the required fraction is $\frac{7}{8}$.

192. The difference between 3/8 and another smaller number is 1/5. Find another number.

(a) $\frac{3}{40}$ (b) $\frac{2}{3}$

(c) $\frac{7}{40}$ (d) $\frac{8}{15}$

RRB Group-D – 06/12/2018 (Shift-II)

Ans. (c) : Let the another smaller number is x.

According to the question,

$$\frac{3}{8}-x=\frac{1}{5}$$

$$-x=\frac{1}{5}-\frac{3}{8}$$

$$-x=\frac{8-15}{40}$$

$$-x=\frac{-7}{40}$$

$$x=\frac{7}{40}$$

Hence, another smaller number is option (c) = $\frac{7}{40}$.

193. The denominator of a fraction exceeds 5 by its numerator. If the numerator is multiplied by 4 and the denominator is multiplied by 3, then the fraction becomes $\frac{1}{2}$.

(a) $\frac{3}{8}$ (b) $\frac{2}{7}$

(c) $\frac{4}{9}$ (d) $\frac{1}{6}$

RRB Group-D – 27/11/2018 (Shift-III)

Ans. (a)

Let the numerator is x, then the fraction $=\frac{x}{x+5}$

According to the question,

$$\frac{x\times 4}{(x+5)\times 3}=\frac{1}{2}$$

$$\frac{4x}{3x+15}=\frac{1}{2}$$

$$8x = 3x + 15$$

$$5x = 15$$

$$x = 3$$

Hence, the required fraction is $\frac{x}{x+5}=\frac{3}{3+5}=\frac{3}{8}$

194. If 2 is added to the square of a positive fraction is $4\frac{1}{4}$. Find the fraction.

(a) $2\frac{3}{4}$ (b) $1\frac{1}{4}$

(c) $2\frac{1}{4}$ (d) $1\frac{1}{2}$

RRB Group-D – 15/11/2018 (Shift-II)

Ans : (d) Let the fraction is x.

According to the question,

$$x^2+2=4\frac{1}{4}$$

$$x^2+2=\frac{17}{4}$$

$$4x^2+8=17$$

$$4x^2=17-8$$

$$4x^2=9\ ,\ x^2=\frac{9}{4}$$

$$x=\frac{3}{2}=1\frac{1}{2}$$

195. 5/12 of a number is 3/4. What is the number?

(a) $3\frac{1}{5}$ (b) $1\frac{7}{5}$

(c) $1\frac{4}{5}$ (d) $1\frac{5}{16}$

RRB Group-D – 10/12/2018 (Shift-I)

Ans. (c) : Let the number is x.
According to the question,

$x \times \frac{5}{12} = \frac{3}{4}$

$\Rightarrow x = \frac{9}{5}$

$x = 1\frac{4}{5}$

196. The sum of three fractions is 5. One of them is $1\frac{1}{2}$, and the difference of two others is $\frac{3}{4}$. Find the greatest of the three.

(a) $2\frac{1}{4}$ (b) $2\frac{1}{8}$

(c) $2\frac{1}{2}$ (d) $1\frac{7}{8}$

RRB Group-D – 09/10/2018 (Shift-II)

Ans. (b) : Let other fractions are x and y.
According to the question,

$x + y + 1\frac{1}{2} = 5$

$\Rightarrow \quad x + y = \frac{7}{2}$ .....(i)

and $\quad x - y = \frac{3}{4}$ .....(ii)

Subtracting equation (ii) from equation (i),

$x + y = \frac{7}{2}$

$x - y = \frac{3}{4}$

$- \quad + \quad -$

$2y = \frac{7}{2} - \frac{3}{4}$

$2y = \frac{11}{4}$

$y = \frac{11}{8} = 1.4$

Putting the value of y in equation (i),

$x + \frac{11}{8} = \frac{7}{2}$

$x = \frac{7}{2} - \frac{11}{8} = \frac{17}{8} = 2\frac{1}{8} = 2.125$

Hence it is clear that the greatest of the three fractions is $2\frac{1}{8}$.

197. Find the divisor of $\frac{1}{5-2\sqrt{3}}$.

(a) $\frac{(5-2\sqrt{3})}{12}$ (b) $\frac{5+2\sqrt{3}}{13}$

(c) $\frac{5-2\sqrt{3}}{13}$ (d) $\frac{5+2\sqrt{3}}{12}$

RRB Group-D – 05/10/2018 (Shift-III)

Ans. (b) On rationalizing $\frac{1}{5-2\sqrt{3}}$

$= \frac{1}{5-2\sqrt{3}} \times \frac{5+2\sqrt{3}}{5+2\sqrt{3}}$

$= \frac{5+2\sqrt{3}}{25-12} = \frac{5+2\sqrt{3}}{13}$

198. Which fraction lies between $\frac{3}{5}$ and $\frac{7}{8}$?

(a) $\frac{8}{9}$ (b) $\frac{6}{7}$

(c) $\frac{12}{13}$ (d) $\frac{4}{7}$

RRB NTPC 18.01.2017 Shift : 1

Ans : (b) From given fractions,

$\frac{3}{5} = 0.60$, $\frac{7}{8} = 0.875$

From question,

(a) $\frac{8}{9} = 0.888$

(b) $\frac{6}{7} = 0.857$

(c) $\frac{12}{13} = 0.923$

(d) $\frac{4}{7} = 0.571$

Hence, it is clear that 0.857 lies between 0.60 and 0.875.

199. If a rod of length $208\frac{4}{5}$ is cut into equal pieces of length $23\frac{1}{5}$ then the total number of rods obtained is:

(a) 5 (b) 7

(c) 8 (d) 9

RRB ALP & Tec. (31-08-18 Shift-I)

Ans : (d) Total length of the rod $= 208\frac{4}{5} = \frac{1044}{5}$

According to the question,

The number of rods obtained = $\frac{\text{Total length}}{\text{Length of one part}}$

$= \frac{\frac{1044}{5}}{23\frac{1}{5}} = \frac{\frac{1044}{5}}{\frac{116}{5}}$

$= \frac{1044}{5} \times \frac{5}{116} = \frac{1044}{116} = 9$

200. A steel rod of length $20\frac{3}{26}$ is cut out from a rod of length $56\frac{1}{5}$. What then is the remaining length of the rod ?

(a) $36\frac{3}{130}$

(b) $36\frac{1}{130}$

(c) $36\frac{11}{130}$

(d) $36\frac{7}{130}$

RRB ALP & Tec. (30-08-18 Shift-II)

Ans : (c) The cut out length of the rod $= 20\frac{3}{26}$

$= \frac{523}{26}$

Total length of the rod $= 56\frac{1}{5} = \frac{281}{5}$

The length of the remaining rod $= \frac{281}{5} - \frac{523}{26}$

$= \frac{7306-2615}{130} = \frac{4691}{130} = 36\frac{11}{130}$

201. If $\frac{3}{4}$ of the weight of a brick is $\frac{7}{8}$ kg, then $\frac{5}{7}$ of the weight of the brick will be:

(a) $\frac{20}{21}$ kg (b) $\frac{5}{6}$ kg

(c) $\frac{5}{8}$ kg (d) $\frac{15}{32}$ kg

RRB ALP & Tec. (14-08-18 Shift-I)

Ans : (b) Let the weight of the brick is x kg.

According to the question,

$\frac{3x}{4} = \frac{7}{8}$

$x = \frac{7}{8} \times \frac{4}{3}$

Hence $\frac{5x}{7} = \frac{7}{8} \times \frac{4}{3} \times \frac{5}{7}$

$\frac{5x}{7} = \frac{5}{6}$

202. Tapan, Ravi and Trisha shared a cake. Tapan had $\frac{1}{4}$ of it, Trisha had $\frac{2}{3}$ of it and Ravi had the rest. What was Ravi's share of the cake?

(a) $\frac{4}{7}$ (b) $\frac{1}{12}$

(c) $\frac{1}{6}$ (d) $\frac{2}{6}$

RRB ALP & Tec. (14-08-18 Shift-II)

Ans : (b) Tapan's share $= \frac{1}{4}$

Trisha's share $= \frac{2}{3}$

Hence, Ravi's share $= 1 - \left(\frac{1}{4} + \frac{2}{3}\right) = 1 - \frac{11}{12} = \frac{1}{12}$

Hence, Ravi's share of the cake is option (b) = $\frac{1}{12}$.

203. The product of two numbers is 0.324. One of the numbers is 1.2. What is the other number?

(a) 2.7 (b) 0.27

(c) 0.027 (d) 27

RRB ALP & Tec. (13-08-18 Shift-II)

Ans : (b) Let the other number is x.

The first number = 1.2

And product = 0.324

According to the question,

$x \times 1.2 = 0.324$

$\boxed{x = \frac{0.324}{1.2} = 0.27}$

204. Tapas, Avi and Rishi shared a cake. Tapas had $\frac{1}{2}$ of it, Rishi had $\frac{1}{3}$ of it and Avi had the rest. What was Avi's share of the cake?

(a) $\frac{2}{6}$ (b) $\frac{1}{6}$

(c) $\frac{3}{6}$ (d) $\frac{3}{5}$

RRB ALP & Tec. (09-08-18 Shift-I)

Ans : (b) Avi's share of the cake,

$= 1 - \left(\frac{1}{2} + \frac{1}{3}\right) = 1 - \left(\frac{3+2}{6}\right)$

$= \frac{6-5}{6} = \frac{1}{6}$

Type - 8

205. Sum of numerator and denominator of a fraction is 13. Adding 3 and 9 to the numerator and denominator respectively, the fraction becomes 2/3. What is the product of the numerator and denominator of the original fraction?

(a) 45 (b) 42

(c) 30 (d) 24

RRB JE - 25/05/2019 (Shift-II)

Ans : (b)

Let the numerator is x and the denominator is y.

So, $x + y = 13$ ..... (i)

$$\frac{x+3}{y+9} = \frac{2}{3}$$

$3x + 9 = 2y + 18$

$3x - 2y = 9$ ..... (ii)

From equation (i) and (ii),

$x = 7, y = 6$

So, the required product = $xy = 7 \times 6 = 42$

206. A cake is shared among five friends. Four of them get the share of the cake $\frac{1}{8}, \frac{1}{6}, \frac{5}{12}, \frac{1}{12}$ respectively. What is the 5th's share of the cake?

(a) $\frac{1}{6}$ (b) $\frac{5}{24}$

(c) $\frac{1}{4}$ (d) $\frac{3}{8}$

RRB Group-D – 17/09/2018 (Shift-I)

Ans : (b) The share of the cake what all four get

$$= \frac{1}{8} + \frac{1}{6} + \frac{5}{12} + \frac{1}{12}$$

$$= \left(\frac{8+6}{48}\right) + \frac{6}{12} = \frac{14}{48} + \frac{6}{12} = \frac{7}{24} + \frac{6}{12} = \frac{19}{24}$$

Hence, the 5th's share of the cake $= 1 - \frac{19}{24} = \frac{5}{24}$

207. $\frac{2}{3}, \frac{4}{6}, \frac{6}{9}$ are–

(a) Odd

(b) Irreducible

(c) Equivalent

(d) Alike (equal)

RRB Group-D – 20/09/2018 (Shift-II)

Ans : (c) From given fractions,

$$\frac{2}{3} = 0.\overline{6}$$

$$\frac{4}{6} = 0.\overline{6}$$

$$\frac{6}{9} = 0.\overline{6} \quad \frac{2}{3} = \frac{4}{6} = \frac{6}{9}$$

Hence, it is clear that all the given fractions are Equivalent.

208. A tennis player, won 18 games out of 27 games played. Calculate the games won in terms of decimal.

(a) 0.667

(b) 0.067

(c) 0.50

(d) 0.333

RRB NTPC 29.04.2016 Shift : 2

Ans : (a) The number of games won $= \frac{18}{27} = 0.667$

209. Saniya won 18 games out of 27 games played. Calculate the games lost in terms of decimal.

(a) 0.333

(b) 0.033

(c) 0.50

(d) 0.667

RRB NTPC 12.04.2016 Shift : 1

Ans : (a)

The number of games lost, $= \frac{27-18}{27}$

$$= \frac{9}{27} = \frac{1}{3} = 0.333$$

210. 0.1, 0.9, 0.01, 0.09, ……., 0.009

(a) 0.0001 (b) 0.1010

(c) 0.001 (d) 0.0011

RRB RPF-SI -13/01/2019 (Shift-II)

Ans : (c) Given,

<u>0.1, 0.9</u>, <u>0.01, 0.09</u>, ..., 0.009

So, the missing term is 0.001,

When we make pairs of two- two, then the pattern showed in blank is 0.001, 0.009.

211. If $\frac{60}{75}$ is equivalent to $\frac{4}{x}$ then the value of x is :

(a) 15 (b) 4

(c) 18 (d) 5

RRB ALP & Tec. (17-08-18 Shift-III)

Ans : (d) According to the question,

$$\frac{60}{75} = \frac{4}{x}$$

$$\Rightarrow \quad x = \frac{4 \times 75}{60}$$

$$\Rightarrow \quad x = 5$$

Hence, the value of x is 5.